Useful Memory under Drift

Temporal-Validity Horizons for Finite-Memory Distribution Tracking

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Abstract

We study the temporal validity of retained evidence for distribution tracking under drift. Longer memory reduces finite-sample noise but also ages the evidence, producing the horizon law

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^H,$$

with optimal scale $n_\star(a, H) \asymp (C_K/\zeta)^{1/(a+H)}$. This law induces a useful-memory region of near-optimal window lengths whose normalized geometry depends only on (a, H). In a tractable one-dimensional root-n proof model we establish the upper law, and a Gaussian lower-bound construction shows that the horizon is structural rather than estimator-specific. For finite families of one-dimensional projections, including coordinate-averaged product Wasserstein, the same root-n horizon law holds under uniform projected control. An inferable-horizon layer then propagates lag-geometry uncertainty into memory selection: when lag geometry is estimated from noisy transport discrepancies, plug-in horizon uncertainty is governed by an operational identifiability criterion and yields a quadratic regret law for memory selection. Experiments show an interior U-curve, roughness-dependent scaling, a real-stream retention signature, and a positive lag between validity loss and standard changepoint alarms.

Keywords: tracking under drift; temporal validity; finite-memory tracking; distribution tracking; Wasserstein distance; Sinkhorn divergence.
MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

In stationary statistics, memory is an unambiguous resource: larger samples reduce variance because the target distribution does not change. Under drift, that logic fails. In streaming estimation, monitoring, and online adaptation, the evidence retained in memory was generated by distributions that may no longer match the current state. More memory therefore does two things at once: it reduces finite-sample noise, and it increases the age of the evidence used to estimate the current state.

That tension creates a horizon problem. Short windows are fresh but noisy. Long windows are stable but stale. The temporal-validity horizon asks how long retained evidence remains useful for estimating the present before drift-induced misalignment overtakes the remaining variance gain.

A general horizon law captures this problem. The law takes the form

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^H,$$

where $a > 0$ is the finite-sample rate exponent of the chosen geometry and estimator, $H \in (0, 1]$ is the temporal roughness exponent of the drift path, and ζ is the roughness amplitude. Balancing these terms yields the temporal-validity horizon

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The horizon depends jointly on finite-sample geometry and on the temporal regularity with which drift accumulates staleness.

At any fixed tolerance $\delta > 0$, the same envelope induces a useful-memory region: the memory lengths whose total tracking error remains within $(1 + \delta)$ times the optimum. The horizon fixes the scale of that region, and its normalized shape depends only on the exponents a and H.

Operational use adds a second question. A horizon can exist structurally while still being statistically difficult to locate from noisy lagged transport discrepancies. The inferable-horizon layer makes that bridge explicit: lag-geometry information must exceed the curvature scale of the useful-memory region for data-driven memory selection to be reliable.

The tractable one-dimensional bounded-support fixed-span triangular-array model establishes the law through the quantile representation of W_2^2, a triangular-array Bahadur expansion, and a design-effect comparison. In that regime, the root-n finite-sample rate

$$\mathbb{E}\, W_2\Big(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\Big) = O(n^{-1/2}).$$

under explicit compact-support, fixed-span, and integrated-remainder hypotheses. This establishes the horizon law in a tractable one-dimensional bounded-support fixed-span model.

The structural lower bound shows that the horizon is imposed by the drift geometry itself. A Gaussian lower-bound construction sharpens the constant-level geometry, and in the Gaussian location model the same object yields a full minimax benchmark on deterministic H"older paths.

For fixed-ε Sinkhorn geometry on an embedded fixed-span model, the proved structural ingredients yield pointwise and bandwise one-dimensional fixed-ε inheritance results on compact support, while the embedded low-dimensional extension isolates the remaining operational frontier.

Broader regular distribution families and operational geometries support different finite-sample rates, and therefore different members of the horizon family. Memory under drift is governed by the interaction between finite-sample geometry and temporal roughness. The compact-support one-dimensional proof model, the structural Gaussian lower bound, and the Gaussian location benchmark close the core law, while finite projection families, coordinate-averaged product Wasserstein, and sliced Wasserstein give multivariate root-n transfer routes under uniform projected control, and fixed-ε Sinkhorn supplies an operational geometry in which the same horizon logic remains visible.

The temporal-validity horizon and the induced useful-memory region are the governing structural objects for finite-memory tracking under drift. The law is proved in a tractable one-dimensional root-n model and matched with structural Gaussian lower theory. Temporal validity and changepoint evidence emerge as distinct statistical clocks, with validity capable of failing before alarm. An inferable-horizon layer then propagates lag-geometry uncertainty into plug-in horizon coverage and operational regret.

The finite-memory U-curve, roughness-dependent horizon scaling, the fixed-ε Sinkhorn evidence, and the measurable lag between validity loss and the first detector alarm are visible in the experiments. Temporal validity and changepoint evidence are different statistical objects: retained evidence can stop representing the present before change becomes detectable. The neighboring literatures on regret, window adaptation, smoothing, filtering, and transport geometry isolate separate pieces of that story; the analysis here identifies the time scale on which retained evidence remains temporally valid before age becomes the dominant source of error.

Section 2 develops the horizon law, useful-memory region, proof model, lower theory, and benchmark results. Section 3 records the main empirical signatures. Section 4 gives the plug-in horizon bridge, and Sections 5 to 8 close the operational frontier, limitations, literature boundary, and main conclusions.

2 Useful Memory and Temporal-Validity Horizons

For a finite-memory estimator with lag weights $w_0, \dots, w_{n-1}$, the natural age variable is the effective lag

$$A(w) := \sum_{j=0}^{n-1} j\, w_j .$$

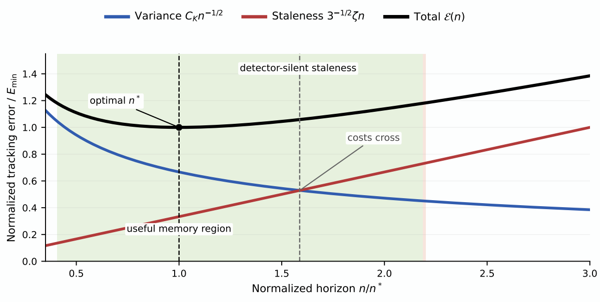

Figure 1: Finite-sample–staleness trade-off under drift. The finite-sample term falls with horizon, the staleness term rises with age, and total error is minimized at the temporal-validity horizon n^*. The green band marks a near-optimal useful-memory region, and the red band marks pre-detection validity loss: retained evidence can stop representing the present before enough changepoint evidence accumulates to trigger an alarm.

Its corresponding staleness at time t is the weighted transport misalignment

$$S_t(w) := \sum_{j=0}^{n-1} w_j\, W_2(P_{t-j}, P_t) .$$

For a uniform window, $A(w) = (n-1)/2$, so window length and temporal age are the same object up to constants.

Non-uniform weights separate two summaries. The finite-sample term is governed by the effective sample size

$$n_{\text{eff}}(w) := \Big(\sum_{j=0}^{n-1} w_j^2 \Big)^{-1},$$

while the drift term is governed by weighted lag moments. For each $H \in (0,1]$, define

$$M_H(w) := \sum_{j=0}^{n-1} j^H w_j .$$

Then any weighted horizon law of the form

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(w)}, P_t\Big) \le C_K n_{\text{eff}}(w)^{-a} + C_S \zeta M_H(w)$$

induces a weighted useful-memory region by balancing the two summaries. On a linear ramp, $M_1(w) = A(w)$ exactly, so the effective lag recovers the exact staleness term.

Proposition 2.1 (Uniform weights maximize effective sample size)**.** *Let $w_0, \dots, w_{n-1}$ be nonnegative weights with $\sum_{j=0}^{n-1} w_j = 1$. Then*

$$n_{\text{eff}}(w) \le n,$$

with equality if and only if $w_j = 1/n$ for all j. Moreover, if the drift profile is an exact linear ramp in the sense that the lag-j contribution is proportional to j, then the weighted staleness equals the effective lag:

$$S_t(w) = c\, A(w)$$

for the corresponding linear coefficient c.

Proof. By Cauchy–Schwarz,

$$1 = \Big(\sum_{j=0}^{n-1} w_j\Big)^2 \le n \sum_{j=0}^{n-1} w_j^2,$$

so $n_{\mathrm{eff}}(w) = 1/\sum_j w_j^2 \le n$, with equality exactly when the weights are all equal. On an exact linear ramp, the lag-j contribution is proportional to j, so averaging with weights w_j gives the weighted mean lag $A(w)$ and hence $S_t(w) = cA(w)$.

Throughout, P_t denotes the present target law, $\bar{P}_t^{(n)}$ the window target obtained by averaging the last n target laws, and $\widehat{P}_t^{(n)}$ the estimator built from the last n observations. The finite-sample term always refers to the estimation error relative to $\bar{P}_t^{(n)}$, while the staleness term refers to the discrepancy between $\bar{P}_t^{(n)}$ and P_t.

Finite-sample error relative to $\bar{P}_t^{(n)}$ and staleness relative to P_t combine into a horizon law. When both sides are controlled, the same scale induces a useful-memory region: the range of memory lengths whose tracking error remains near optimal.

2.1 General Horizon Law

The main theorem separates finite-sample behavior from drift-induced staleness.

Theorem 2.2 (Abstract upper law)**.** *Let d be a probability metric satisfying the triangle inequality. At time t, let P_t be the present target law, let $\bar{P}_t^{(n)}$ be the window target, and let $\widehat{P}_t^{(n)}$ be the estimator built from the last n observations. Assume that for some constants $C_K > 0$, $C_S > 0$, exponent $a > 0$, roughness index $H \in (0, 1]$, and amplitude $\zeta > 0$,*

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) \le C_K n^{-a}$$

and

$$d\Big(\bar{P}_t^{(n)}, P_t\Big) \le C_S \zeta n^H$$

hold for all n in the regime of interest. Then

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^H.$$

Proof. By the triangle inequality,

$$d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le d\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) + d\Big(\bar{P}_t^{(n)}, P_t\Big).$$

Taking expectations and applying the two assumptions yields the claim.

The theorem isolates the structure of the problem: the path class supplies the staleness term, while the chosen geometry and sample regime supply the finite-sample term.

Proposition 2.3 (Triangular-array transfer for Hadamard-differentiable functionals)**.** *Let $\mathcal{T}$ be a functional on probability laws that is Hadamard differentiable at P_t tangentially to a linear subspace of signed measures, with derivative $\dot{\mathcal{T}}_{P_t}$ bounded on the relevant tangent class. Suppose that for the triangular-array estimator and its window target, the base metric satisfies*

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) \le C_K n^{-a}$$

and the drift envelope satisfies

$$d\Big(\bar{P}_t^{(n)}, P_t\Big) \le C_S \zeta n^H.$$

If the remainder in the Hadamard expansion is $o(n^{-a})$ uniformly on the relevant class, then there exist constants $\widetilde{C}_K, \widetilde{C}_S > 0$ such that

$$\mathbb{E}\,\big\|\mathcal{T}(\widehat{P}_t^{(n)}) - \mathcal{T}(P_t)\big\| \le \widetilde{C}_K n^{-a} + \widetilde{C}_S \zeta n^H.$$

Proof. Write the Hadamard expansion at P_t and apply the boundedness of the derivative to the centered triangular-array fluctuation. The drift term contributes through the same envelope as the base metric because $\mathcal{T}$ is locally Lipschitz on the class. The remainder is lower order by assumption, so the transformed law inherits the same horizon envelope.

Proposition 2.4 (Finite-dimensional smooth transfer)**.** *Let $\phi_1, \dots, \phi_k$ be bounded measurable functions and let $Z(P) = \big(\mathbb{E}_P[\phi_1], \dots, \mathbb{E}_P[\phi_k]\big)$. Let $g \in C^2(U)$ on an open set $U \subset \mathbb{R}^k$ containing the relevant range of Z. Define $T(P) = g(Z(P))$. Suppose that on the triangular-array window target and estimator,*

$$\mathbb{E}\,\|Z(\widehat{P}_t^{(n)}) - Z(\bar{P}_t^{(n)})\| \le C_K n^{-a}$$

and

$$\|Z(\bar{P}_t^{(n)}) - Z(P_t)\| \le C_S \zeta n^H.$$

If the Hessian of g is bounded by M on the relevant range, then

$$\mathbb{E}\,|T(\widehat{P}_t^{(n)}) - T(P_t)| \le \widetilde{C}_K n^{-a} + \widetilde{C}_S \zeta n^H,$$

and the Taylor remainder is quadratic in the moment error:

$$\begin{aligned} &|T(P') - T(P) - \nabla g(Z(P))^\top \big(Z(P') - Z(P)\big)| \\ &\quad \le \frac{M}{2}\|Z(P') - Z(P)\|^2. \end{aligned}$$

Proof. Apply the multivariate Taylor formula to g around $Z(P_t)$. The bounded Hessian gives the stated quadratic remainder. The first-order term is controlled by the assumed finite-sample and drift envelopes for Z, and the same decomposition as in proposition 2.3 yields the horizon law for T.

The same decomposition extends to non-uniform weights once the two summaries $n_{\mathrm{eff}}(w)$ and $M_H(w)$ are controlled. The theorem-level question is whether weighted forgetting enlarges the useful-memory region beyond a constant factor on richer path classes, or whether it only changes the leading constant.

Corollary 2.5 (Temporal-validity horizon)**.** *Under the assumptions of theorem 2.2, the upper envelope*

$$\Phi(n) := C_K n^{-a} + C_S \zeta n^H$$

is minimized, up to constant factors depending only on a, H, and C_S, at

$$n^*(a,H) \asymp (C_K/\zeta)^{1/(a+H)},$$

and the corresponding error scale is

$$\Phi(n^*) \asymp C_K^{H/(a+H)} \zeta^{a/(a+H)}.$$

Proof. Balance the decreasing finite-sample term against the increasing staleness term:

$$C_K n^{-a} \asymp C_S \zeta n^H.$$

This gives $n^{a+H} \asymp C_K/\zeta$, hence the displayed horizon law. Substituting that scale back into either term gives the optimal error scale.

Write the continuous optimizer of Φ as

$$n_\star := \left(\frac{aC_K}{HC_S\zeta}\right)^{1/(a+H)}.$$

This optimizer differs from the scale $n^*(a,H)$ only by constants depending on a, H, and C_S, so the asymptotic horizon law remains unchanged.

2.2 Useful-Memory Region

The horizon scale induces a near-optimal region of memory lengths. For any $\delta > 0$, define the useful-memory region by

$$\mathcal{U}_\delta := \{n > 0 : \Phi(n) \le (1+\delta)\Phi(n_\star)\}.$$

This object records not only where the envelope is minimized, but also how much memory remains acceptable up to a fixed tolerance.

Proposition 2.6 (Useful-memory region at tolerance δ)**.** *Let $\Phi(n) = C_K n^{-a} + C_S\zeta n^H$ with $a, H > 0$, and let $n_\star$ be its continuous optimizer. Then the normalized profile*

$$\Psi(x) := \frac{\Phi(n_\star x)}{\Phi(n_\star)} = \frac{Hx^{-a} + ax^H}{a+H}$$

is strictly decreasing on $(0,1]$ and strictly increasing on $[1,\infty)$. Consequently,

$$\mathcal{U}_\delta = [n_-, n_+] = n_\star[x_-(\delta;a,H), x_+(\delta;a,H)],$$

where $x_-(\delta;a,H) \le 1 \le x_+(\delta;a,H)$ are the two solutions of $\Psi(x) = 1+\delta$. In particular, the useful-memory region scales linearly with the horizon scale $n_\star$, and its relative shape depends only on a, H, and δ.

Proof. Since $n_\star$ minimizes Φ, differentiation gives

$$aC_K n_\star^{-a} = HC_S\zeta n_\star^H.$$

Substituting $n = n_\star x$ into Φ and dividing by $\Phi(n_\star)$ yields the displayed identity for Ψ. Differentiation then gives

$$\Psi'(x) = \frac{aH}{a+H}\big(x^{H-1} - x^{-a-1}\big),$$

which is negative for $x < 1$ and positive for $x > 1$. Since $\Psi(1) = 1$ and $\Psi(x) \to \infty$ as $x \to 0$ or $x \to \infty$, the sublevel set $\{x > 0 : \Psi(x) \le 1+\delta\}$ is a compact interval containing 1, which proves the representation of $\mathcal{U}_\delta$.

For small tolerance, the useful-memory region has the local expansion

$$x_\pm(\delta;a,H) = 1 \pm \sqrt{\frac{2\delta}{aH}} + o(\sqrt{\delta}), \qquad \delta \downarrow 0,$$

because $\Psi''(1) = aH$. The local curvature therefore controls how quickly the near-optimal region contracts around the horizon.

The horizon depends jointly on the finite-sample rate exponent and the roughness exponent. Different geometries and different drift regularities induce different memory scales.

2.3 Drift Regularity and Staleness Accumulation

The staleness side of the law is controlled by a deterministic path class. For $H \in (0,1]$ and $\zeta > 0$, write

$$\mathfrak{P}_H(\zeta) := \left\{(P_s)_{s\in\mathbb{Z}} : W_2(P_{t-j}, P_t) \le \zeta j^H \text{ for all } j \ge 0\right\}.$$

The case $H = 1$ is the local Lipschitz envelope; smaller H correspond to slower accumulation of staleness with lag.

Lemma 2.7 (Hölder staleness bound via averaged couplings)**.** *Let*

$$\bar{P}_t^{(n)} := \frac{1}{n}\sum_{j=0}^{n-1} P_{t-j}.$$

If the target path belongs to $\mathfrak{P}_H(\zeta)$, then

$$W_2^2\Big(\bar{P}_t^{(n)}, P_t\Big) \le \frac{\zeta^2}{n}\sum_{j=0}^{n-1} j^{2H},$$

and therefore, by Jensen's inequality,

$$W_2\Big(\bar{P}_t^{(n)}, P_t\Big) \le \zeta\left(\frac{1}{n}\sum_{j=0}^{n-1} j^{2H}\right)^{1/2} = C_{H,n}\,\zeta n^H,$$

where

$$C_{H,n} := \left(n^{-(2H+1)}\sum_{j=0}^{n-1} j^{2H}\right)^{1/2}.$$

In particular, $C_{H,n} \to (2H+1)^{-1/2}$ as $n \to \infty$.

Proof. For each lag j, let π_j be an optimal coupling between P_{t-j} and P_t. Their average

$$\bar{\pi} := \frac{1}{n}\sum_{j=0}^{n-1} \pi_j$$

couples $\bar{P}_t^{(n)}$ with P_t, so

$$\begin{aligned} W_2^2\Big(\bar{P}_t^{(n)}, P_t\Big) &\le \int \|x-y\|^2\, d\bar{\pi}(x,y) \\ &= \frac{1}{n}\sum_{j=0}^{n-1} W_2^2(P_{t-j}, P_t). \end{aligned}$$

Applying the Hölder envelope gives the displayed finite-n constant, and taking square roots completes the proof.

Combining lemma 2.7 with corollary 2.5 yields the horizon family

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

In the root-n finite-sample regime $a = 1/2$,

$$n^*(1/2, H) \asymp (C_K/\zeta)^{2/(1+2H)}.$$

The finite-n constant in lemma 2.7 is the constant used here for uniform windows; the simpler $(2H+1)^{-1/2}$ expression is its large-window limit.

2.4 One-Dimensional Proof Model for the Root-n Regime

The bounded-support fixed-span one-dimensional triangular array provides a tractable rigorous finite-sample instantiation of the abstract law. The model is built on common compact support, uniformly regular window mixtures, and an integrated Bahadur remainder hypothesis.

Proposition 2.8 (Root-n finite-sample bound in the 1D proof model)**.** *Let $X_{n,1}, \dots, X_{n,n}$ be independent one-dimensional observations with laws $P_{n,1}, \dots, P_{n,n}$, and define the window mixture*

$$\bar{P}_n = \frac{1}{n}\sum_{j=1}^{n} P_{n,j}.$$

Assume:

(i) all $P_{n,j}$ are supported on a common compact interval $[L, U]$;

(ii) the within-window drift span is uniformly bounded in n;

(iii) each mixture law $\bar{P}_n$ has a density $\bar{f}_n$ on $[L, U]$ that is bounded below by a positive constant and uniformly Hölder;

(iv) writing q_n for the quantile function of $\bar{P}_n$ and $\widehat{q}_n$ for the empirical quantile function of $\widehat{P}_n^{\mathrm{tri}}$, one has the Bahadur representation

$$\widehat{q}_n(u) - q_n(u) = \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u), \qquad u \in (0,1),$$

with

$$\mathbb{E}\int_0^1 R_n(u)^2\, du = o(n^{-1}).$$

Then

$$\mathbb{E}\, W_2^2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = O(n^{-1}),$$

and therefore, by Jensen's inequality,

$$\mathbb{E}\, W_2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = O(n^{-1/2}).$$

Proof. In one dimension,

$$W_2^2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = \int_0^1 \left(\widehat{q}_n(u) - q_n(u)\right)^2 du,$$

so the problem reduces to the integrated L_2 error of the empirical quantile process. By item (iv),

$$\widehat{q}_n(u) - q_n(u) = \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u).$$

Define

$$Z_n(u) := \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))}.$$

Then

$$\mathbb{E}\int_0^1 \left(\widehat{q}_n(u) - q_n(u)\right)^2 du = \mathbb{E}\int_0^1 \left(Z_n(u) + R_n(u)\right)^2 du.$$

Let $m > 0$ be a uniform lower bound for $\bar{f}_n$. Since

$$\widehat{F}_n(x) = \frac{1}{n}\sum_{j=1}^{n} \mathbf{1}\{X_{n,j} \le x\},$$

independence across the row gives

$$\mathrm{Var}(\widehat{F}_n(q_n(u))) = \frac{1}{n^2}\sum_{j=1}^{n} p_{n,j}(u)(1 - p_{n,j}(u)) \le \frac{1}{4n},$$

where $p_{n,j}(u) = P_{n,j}((-\infty, q_n(u)])$. Therefore

$$\mathbb{E}\int_0^1 Z_n(u)^2 du \le m^{-2}\int_0^1 \mathrm{Var}(\widehat{F}_n(q_n(u)))du \le \frac{1}{4m^2 n}.$$

By item (iv),

$$\mathbb{E}\int_0^1 R_n(u)^2 du = o(n^{-1}).$$

The cross term is then $o(n^{-1})$ by Cauchy–Schwarz, since $\mathbb{E}\int_0^1 Z_n(u)^2 du = O(n^{-1})$ and $\mathbb{E}\int_0^1 R_n(u)^2 du = o(n^{-1})$. Hence the expected squared Wasserstein error is $O(n^{-1})$, and Jensen's inequality gives the root-n rate for $\mathbb{E}\, W_2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right)$.

The main conclusion is the exponent. Under this bounded-support fixed-span regime, the triangular-array finite-sample term remains root-n.

2.5 Verifiable Sufficient Conditions for the 1D Proof Model

An explicit deformation class makes the proof-model hypotheses verifiable. Let

$$g_\delta(x) = x + \delta x(1-x), \qquad x \in [0,1], \qquad |\delta| \le \Delta < 1.$$

Then g_δ is an increasing C^2 diffeomorphism of $[0,1]$, with

$$1 - \Delta \le g_\delta'(x) \le 1 + \Delta, \qquad |g_\delta''(x)| \le 2\Delta,$$

so the transformed densities satisfy

$$\frac{1}{1+\Delta} \le f_\delta(y) \le \frac{1}{1-\Delta},$$

and the same support is preserved.

Corollary 2.9 (Verifiable Bahadur remainder for the smooth deformation class)**.** *Assume that each row law has the form*

$$P_{n,j} = (g_{\delta_{n,j}})_{\#}\,\mathrm{Unif}[0,1], \qquad |\delta_{n,j}| \le \Delta < 1,$$

and that the within-window span is uniformly bounded in n. Then item (iv) of proposition 2.8 holds. More precisely, there exists a constant $C_\Delta < \infty$ such that

$$\mathbb{E}\int_0^1 R_n(u)^2\,du \le C_\Delta n^{-2}.$$

Consequently,

$$\mathbb{E}\, W_2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = O(n^{-1/2})$$

on this deformation class.

Proof. Write $\bar{F}_n$ and q_n for the mixture cdf and quantile. Since each $g_{\delta_{n,j}}$ is an increasing diffeomorphism of $[0,1]$, every mixture law $\bar{P}_n$ is supported on $[0,1]$ and has density $\bar{f}_n = \frac{1}{n}\sum_{j=1}^n f_{\delta_{n,j}}$ satisfying the explicit bounds

$$\frac{1}{1+\Delta} \le \bar{f}_n(y) \le \frac{1}{1-\Delta}, \qquad \|\bar{f}_n'\|_\infty \le \frac{2\Delta}{(1-\Delta)^3}.$$

Hence the inverse map $\bar{F}_n^{-1}$ is uniformly twice differentiable on $(0,1)$, with second derivative bounded by a constant depending only on Δ. Applying the second-order Taylor expansion of $\bar{F}_n^{-1}$ around $u = \bar{F}_n(q_n(u))$ gives

$$\widehat{q}_n(u) - q_n(u) = \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u),$$

with

$$|R_n(u)| \le C_\Delta\, |\widehat{F}_n(q_n(u)) - u|^2.$$

Now

$$\widehat{F}_n(q_n(u)) = \frac{1}{n}\sum_{j=1}^n \mathbf{1}\{X_{n,j} \le q_n(u)\}$$

is an average of independent Bernoulli variables with mean u. Uniformly in u, its fourth centered moment is $O(n^{-2})$. Therefore

$$\mathbb{E}\, R_n(u)^2 \le C_\Delta\, \mathbb{E}|\widehat{F}_n(q_n(u)) - u|^4 \le C_\Delta' n^{-2},$$

and integrating over $u \in (0,1)$ yields

$$\mathbb{E}\int_0^1 R_n(u)^2\,du \le C_\Delta' n^{-2} = o(n^{-1}).$$

The root-n claim then follows from proposition 2.8. The same smoothness package is consistent with the classical Bahadur theory of Bahadur [1966] and van der Vaart [1998, Chapter 21], but here the explicit deformation bounds make the integrated remainder quantitative.

Remark 2.10 (Weak rowwise dependence). The proof of proposition 2.8 uses independence within a row only through the $O(n^{-1})$ variance bound for the empirical cdf term and through item (iv). Consequently, the same root-n conclusion persists under any short-range dependent rowwise model for which the empirical cdf still has L_2 fluctuation of order $n^{-1/2}$ and the integrated Bahadur remainder remains $o(n^{-1})$. Short-range mixing conditions are one natural route to such a package, but the present paper does not develop a general dependent-row theorem.

On the tested smooth fixed-support family, the integrated residual decays at empirical rates between 1.40 and 1.54 for spans 0.2 and 0.4, and the scaled quantity $n\int_0^1 R_n(u)^2 du$ still decays at rates between 0.40 and 0.54; see appendix B. On the fixed-span triangular-array uniform-kernel model, the residual-to-MSE ratio drops from about 0.38 at $n = 25$ to about 0.11 at $n = 400$, with integrated residual rates 1.43 for the triangular sample and 1.49 for the i.i.d. mixture benchmark. These diagnostics do not replace a general verification theorem, but they sit behind an explicit sufficient-condition corollary on the smooth deformation class and quantify span dependence on the tested classes.

2.6 Structural Lower Bound in the Root-n Regime

The upper law would be substantially weaker if it were only an estimator-specific statement. The lower bound shows that, in the root-n finite-sample regime $a = 1/2$, the horizon is imposed by the drift geometry itself.

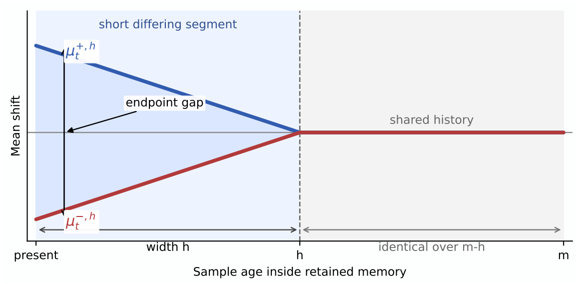

Figure 2: Gaussian lower-bound construction. The two paths share most of the retained history and differ only on a recent segment of width h, yet remain separated at the present. That recent segment already produces the same horizon exponent as the upper law in the root-n regime.

Proposition 2.11 (Structural lower bound in the root-n regime)**.** *Fix $H \in (0,1]$ and work on the Gaussian location subclass with known variance σ^2 and path pair*

$$\mu_{t-j}^{\pm,h} = \pm\beta(h-j)_+^H, \qquad j = 0,1,\dots,h,$$

where $\beta \le \zeta$ enforces the roughness budget. Let $\mathcal{G}_{H,\zeta}^{\mathrm{Gauss}}$ denote the corresponding Gaussian location subclass of target paths, and suppose the retained observations are independent with laws $N(\mu_{t-j}^{\pm,h}, \sigma^2)$. Then there exists a constant $c_H > 0$ depending only on H such that the endpoint tracking risk over this subclass satisfies

$$\inf_{\widehat{P}}\ \sup_{P \in \mathcal{G}_{H,\zeta}^{\mathrm{Gauss}}} \mathbb{E}\, W_2\left(\widehat{P}_t, P_t\right) \ge c_H\, \sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}.$$

Consequently, in the root-n finite-sample regime $a = 1/2$, the horizon scale

$$h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$$

is structural even on this restricted subclass.

Proof. Compare two hypotheses generated by the path pair $\mu_{t-j}^{+,h}$ and $\mu_{t-j}^{-,h}$. At the present time, the endpoint separation is of order βh^H. On the Gaussian location subclass, this is also the W_2 separation. The Kullback–Leibler divergence between the two hypotheses is proportional to

$$\beta^2 \sum_{r=1}^{h} r^{2H}/\sigma^2.$$

Choosing

$$\beta = \min\left\{\zeta, \frac{\sigma}{2\sqrt{\sum_{r=1}^{h} r^{2H}}}\right\}$$

keeps the testing problem in the Le Cam regime while respecting the roughness budget. The resulting two-point lower bound is of order βh^H. Balancing the active roughness constraint against the testing constraint gives $h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$ and therefore the displayed lower law. The exact Gaussian two-point calculation used here is recorded in appendix A.

The lower bound is subclass based and not class-tight, but it is enough to show that the horizon is a structural feature of the problem and not an estimator artifact.

The proposition above is the lower-bound fact needed for the main horizon law. The same Gaussian construction also exposes the constant-level geometry. The following proposition records the sharp Gaussian constants available within that proof scheme.

2.7 Gaussian Lower-Bound Geometry

Let Φ and φ denote the standard normal distribution function and density.

Proposition 2.12 (Gaussian lower-bound constants)**.** *Fix $H \in (0,1]$ and set $p_H := 2H/(2H+1)$. Let $x_H > 0$ be the unique maximizer of $x^{p_H}\Phi(-x)$, equivalently the unique solution of*

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

For the Gaussian ramp profile $\mu_{t-j}^{\pm,h} = \pm\zeta(h-j)_+^H$, let

$$L_h^{\mathrm{ramp}}(\sigma,\zeta) := \zeta h^H \Phi\left(-\frac{\zeta}{\sigma}\sqrt{\sum_{r=1}^{h} r^{2H}}\right).$$

Then, as $\sigma/\zeta \to \infty$,

$$\sup_{h\geq 1} L_h^{\mathrm{ramp}}(\sigma,\zeta) \sim C_H^{\mathrm{ramp}}\, \sigma^{2H/(2H+1)}\zeta^{1/(2H+1)},$$

with

$$C_H^{\mathrm{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)}\Phi(-x_H).$$

Among all endpoint-saturating Hölder profiles, the profile

$$g_r^{\min} = h^H - (h-r)^H$$

minimizes the testing energy and has asymptotic energy constant

$$I_H = \int_0^1 \left(1-(1-u)^H\right)^2 du = \frac{2H^2}{(H+1)(2H+1)}.$$

If

$$L_h^{\min}(\sigma,\zeta) := \zeta h^H \Phi\left(-\frac{\zeta}{\sigma}\sqrt{\sum_{r=1}^{h} (g_r^{\min})^2}\right),$$

then, as $\sigma/\zeta \to \infty$,

$$\sup_{h\geq 1} L_h^{\min}(\sigma,\zeta) \sim C_H^{\min}\, \sigma^{2H/(2H+1)}\zeta^{1/(2H+1)},$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)}\Phi(-x_H).$$

For $H < 1$, this strictly improves the ramp constant by the factor

$$\frac{C_H^{\min}}{C_H^{\mathrm{ramp}}} = \left(\frac{H+1}{2H^2}\right)^{H/(2H+1)} > 1,$$

while at $H = 1$ the two profiles coincide.

Proof. The exact Gaussian testing problem reduces the two-point lower bound to the displayed quantities L_h^{ramp} and $L_h^{\min}$. After rescaling by $h = A(\sigma/\zeta)^{2/(2H+1)}$, the ramp energy and endpoint-minimal energy have limits $(2H+1)^{-1}$ and I_H, respectively. The corresponding normalized payoff becomes a scalar multiple of $x^{p_H}\Phi(-x)$, whose unique maximizer is x_H. The endpoint-minimal profile minimizes testing energy among all endpoint-saturating Hölder profiles, so it yields the sharper constant. Full details are given in appendix A.

These constants sharpen the lower-bound picture without changing the horizon exponents or the status of the main results.

The same lower-bound geometry also yields a full class-level benchmark once the model is specialized to Gaussian location tracking. In that setting, the deterministic Hölder envelope, the finite-memory estimator, and the lower-bound construction all live in the same one-dimensional parameterization, so this model admits a direct minimax rate statement.

Theorem 2.13 (Gaussian location minimax benchmark on the deterministic Hölder class)**.** *Fix $H \in (0,1]$, $\zeta > 0$, and $\sigma > 0$. Let $\mathfrak{M}_H(\zeta)$ denote the class of deterministic mean paths $(\mu_s)_{s\in\mathbb{Z}}$ such that*

$$|\mu_{t-j} - \mu_t| \leq \zeta j^H \qquad \text{for all } j \geq 0.$$

Suppose the observations are independent with

$$Y_{t-j} \sim N(\mu_{t-j}, \sigma^2), \qquad j = 0,1,\ldots,n-1,$$

and let the target law at time t be $P_t = N(\mu_t,\sigma^2)$. Define the optimal finite-memory minimax risk

$$\mathcal{R}_H^*(\sigma,\zeta) := \inf_{n\geq 1}\inf_{\widehat{P}_t^{(n)}}\sup_{\mu\in\mathfrak{M}_H(\zeta)} \mathbb{E}\, W_2\left(\widehat{P}_t^{(n)}, P_t\right).$$

Then there exist constants $0 < c_H \leq C_H < \infty$, depending only on H, such that

$$\begin{aligned} c_H\, \sigma^{2H/(2H+1)}\zeta^{1/(2H+1)} &\leq \mathcal{R}_H^*(\sigma,\zeta) \\ &\leq C_H\, \sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}. \end{aligned}$$

In particular, the deterministic Hölder class has horizon scale

$$n_H^* \asymp (\sigma/\zeta)^{2/(2H+1)}.$$

Moreover, the uniform-window mean estimator has an exact asymptotic upper constant on this benchmark. If $y_H > 0$ solves

$$Hy_H\big(2\Phi(y_H) - 1\big) = \varphi(y_H),$$

then its asymptotic horizon shape is

$$A_H^{\mathrm{mean}} = ((H+1)y_H)^{2/(2H+1)}$$

and its asymptotic risk constant is

$$U_H^{\mathrm{mean}} = ((H+1)y_H)^{-1/(2H+1)} \times \Big(2\varphi(y_H) + y_H\big(2\Phi(y_H) - 1\big)\Big).$$

Proof. For the upper bound, estimate the present mean by the uniform-window average

$$\widehat{\mu}_t^{(n)} := \frac{1}{n}\sum_{j=0}^{n-1} Y_{t-j}$$

and set $\widehat{P}_t^{(n)} = N(\widehat{\mu}_t^{(n)}, \sigma^2)$. Because equal-variance Gaussian laws satisfy

$$W_2\big(N(m_1,\sigma^2), N(m_2,\sigma^2)\big) = |m_1 - m_2|,$$

the risk reduces to mean estimation error. The noise term is $\sigma\sqrt{2/(\pi n)}$, while the deterministic bias is bounded by

$$\frac{\zeta}{n}\sum_{j=0}^{n-1} j^H,$$

so

$$\sup_{\mu\in\mathfrak{M}_H(\zeta)} \mathbb{E}\, W_2\Big(\widehat{P}_t^{(n)}, P_t\Big) \le \sigma\sqrt{\frac{2}{\pi n}} + \frac{\zeta}{n}\sum_{j=0}^{n-1} j^H.$$

Rescaling by $n = A(\sigma/\zeta)^{2/(2H+1)}$ yields a one-parameter normalized upper envelope. Differentiating its exact Gaussian-location form gives the stated root equation for y_H, and therefore the displayed asymptotic upper constant and horizon shape. In particular, this proves the stated upper rate and horizon scale.

For the lower bound, use the symmetric two-point subfamily

$$\mu_{t-j}^{\pm,h} = \pm\beta\big(h^H - j^H\big)_+, \qquad j \ge 0,$$

with $\beta \le \zeta$. This family lies in $\mathfrak{M}_H(\zeta)$ because

$$|\mu_t^{\pm,h} - \mu_{t-j}^{\pm,h}| = \beta\min(h^H, j^H) \le \zeta j^H.$$

Moreover, for fixed endpoint value βh^H, it minimizes the Gaussian testing energy over all symmetric lag profiles allowed by the present-point Hölder envelope, since each lag coordinate is independently minimized by $\beta(h^H - j^H)_+$. The exact Gaussian two-point calculation from the lower-bound analysis in appendix A therefore applies on the full deterministic Hölder class and yields a lower bound of order

$$\sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}.$$

Minimizing over h gives the same horizon scale $(\sigma/\zeta)^{2/(2H+1)}$. Combining the two sides proves the theorem.

2.8 Multivariate Extension via Finite Projection Families, Sliced, and Product Geometries

The horizon law also extends beyond one dimension when the multivariate geometry reduces to a uniformly controlled family of one-dimensional transport problems. This is the natural setting for sliced or coordinate-averaged Wasserstein constructions, where the finite-sample term is built by integrating projected one-dimensional errors.

Corollary 2.14 (Finite-family projection transfer)**.** *Let $\Pi = \{\theta_1,\dots,\theta_m\}$ be a finite family of linear maps from the ambient space to $\mathbb{R}$, and define*

$$d_\Pi(P,Q) := \Big(\frac{1}{m}\sum_{r=1}^{m} W_2^2(P_{\theta_r}, Q_{\theta_r})\Big)^{1/2}.$$

Assume that for every $r = 1,\dots,m$:

(i) the projected triangular-array window satisfies the hypotheses of proposition 2.8 with constants uniform in r;

(ii) the projected drift path obeys the same Hölder staleness envelope with exponent H and amplitude at most ζ uniformly in r.

Then there exist constants $C_{K,\Pi}, C_{S,\Pi} < \infty$ such that

$$\mathbb{E}\, d_\Pi\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_{K,\Pi} n^{-1/2} + C_{S,\Pi}\zeta n^H.$$

Consequently,

$$n_\Pi^*(H) \asymp (C_{K,\Pi}/\zeta)^{1/(H+1/2)}.$$

In particular, the same law holds for coordinate-averaged product constructions obtained by taking Π to be the coordinate projections.

Proof. For each r, proposition 2.8 gives

$$\mathbb{E}\, W_2^2\Big(\widehat{P}_{t,\theta_r}^{(n)}, \bar{P}_{t,\theta_r}^{(n)}\Big) \le Cn^{-1}$$

with C uniform in r. Averaging over r yields

$$\mathbb{E}\, d_\Pi^2\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) \le Cn^{-1},$$

and Jensen gives the root-n bound for $\mathbb{E}\, d_\Pi(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)})$. The projected Hölder bound averages in the same way:

$$d_\Pi\Big(\bar{P}_t^{(n)}, P_t\Big) \le C_{S,\Pi}\zeta n^H.$$

Applying theorem 2.2 with $d = d_\Pi$ yields the displayed envelope and horizon scale.

Corollary 2.15 (Sliced-Wasserstein transfer under uniform projected control)**.** *Let ρ be a probability measure on a family of one-dimensional projections θ, and define*

$$SW_{2,\rho}^2(P,Q) := \int W_2^2(P_\theta, Q_\theta)\, \rho(d\theta),$$

where P_θ denotes the pushforward of P under projection θ. Assume that for every θ in the support of ρ:

(i) the projected triangular-array window satisfies the hypotheses of proposition 2.8 with constants uniform in θ;

(ii) the projected drift path obeys the same Hölder staleness envelope with exponent H and amplitude at most ζ uniformly in θ.

Then there exist constants $C_K, C_S < \infty$ such that

$$\mathbb{E}\, SW_{2,\rho}\left(\widehat{P}_t^{(n)}, P_t\right) \le C_K n^{-1/2} + C_S \zeta n^H.$$

Consequently,

$$n^*(H) \asymp (C_K/\zeta)^{1/(H+1/2)}.$$

Finite projection families are included by taking ρ supported on finitely many atoms; corollary 2.14 states that case directly for multivariate tracking.

Proof. For each projection θ, proposition 2.8 gives

$$\mathbb{E}\, W_2^2\left(\widehat{P}_{t,\theta}^{(n)}, \bar{P}_{t,\theta}^{(n)}\right) \le Cn^{-1}$$

with C uniform in θ. Integrating over ρ yields

$$\mathbb{E}\, SW_{2,\rho}^2\left(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) \le Cn^{-1},$$

and Jensen gives the root-n bound for $\mathbb{E}\, SW_{2,\rho}(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)})$. The projected Hölder bound integrates in the same way, giving

$$SW_{2,\rho}\left(\bar{P}_t^{(n)}, P_t\right) \le C_S \zeta n^H.$$

Applying theorem 2.2 with $d = SW_{2,\rho}$ yields the stated envelope and horizon scale.

2.9 Benchmark Theorems and Operational Extensions

The Gaussian benchmark and the fixed-ε Sinkhorn extension place the horizon law in two broader settings: a full path-class benchmark with theorem- level closure, and an operational transport geometry in which the one- dimensional compact-support inheritance theory is already explicit. The linear-Gaussian entropic-OT setting suggests an intermediate recursive regime between the i.i.d. fixed-ε theory and the embedded low-dimensional extension [Kalman, 1960, Jazwinski, 1970, Akyildiz et al., 2024].

Fixed-ε Sinkhorn setting. For fixed-ε debiased Sinkhorn, the central question is whether a practical geometry can supply a stable finite-sample term on the embedded fixed-span model. The target law is

$$\mathbb{E}\, S_\varepsilon\left(\widehat{P}_{t,\mathrm{tri}}^{(n)}, \bar{P}_t^{(n)}\right) \le C_\varepsilon n^{-a_\varepsilon}$$

under bounded support, fixed span, and structural low-dimensionality, where S_ε denotes a fixed-ε Sinkhorn-type geometry and the exponent a_ε matches the corresponding i.i.d. mixture benchmark.

Definition 2.16 (Fixed-ε Sinkhorn setting)**.** Fix a bounded embedded fixed-span model, an intrinsic complexity index k, a dual Hölder smoothness level α, and an operational band $\mathcal{B} \subset (0, \infty)$ of regularization values. We say that this model is in the fixed-ε Sinkhorn setting if:

(i) the i.i.d. fixed-ε benchmark on the same support class admits a finite-sample bound with exponent a_ε on $\mathcal{B}$;

(ii) the triangular-window and i.i.d.-mixture supports have identical operational covering complexity on $\mathcal{B}$;

(iii) the dual class lies in the parametric adaptation region $\alpha > k/2$.

Proposition 2.17 (Structural ingredients for fixed-ε Sinkhorn)**.** *On the bounded embedded fixed-span model with squared-Euclidean cost, the first three items in definition 2.16 are structural: the support-complexity inheritance is exact, and the dual-complexity condition is precisely the LCA threshold $\alpha > k/2$.*

Proposition 2.18 (Conditional embedded moderate-band Sinkhorn inheritance)**.** *Fix the calibrated embedded fixed-span model with intrinsic dimension $k = 2$ and a compact moderate band $\mathcal{B} = [\varepsilon_{\min}, \varepsilon_{\max}]$ bounded away from zero. Assume:*

(i) a centered self-coupling stability lemma holds uniformly on $\mathcal{B}$ with a spectral-radius bound strictly below one and a uniform inverse-norm bound on the centered subspace;

(ii) the centered Sinkhorn scaling map admits a uniform implicit-function linearization on $\mathcal{B}$ with remainder $o(n^{-1/2})$;

(iii) the induced influence class satisfies a root-n triangular empirical- process bound on the embedded support class;

(iv) the embedded fixed-span support and dual-complexity hypotheses of proposition 2.17 hold.

Then there exists a finite constant $C_{\mathcal{B}}$ such that

$$\sup_{\varepsilon\in\mathcal{B}} \mathbb{E}\, S_\varepsilon\left(\widehat{P}_{t,\mathrm{tri}}^{(n)}, \bar{P}_t^{(n)}\right) \le C_{\mathcal{B}} n^{-1/2}.$$

Consequently the corresponding temporal-validity horizon on S_ε has the root-n horizon scale

$$n_{\mathcal{B}}^*(H) \asymp (C_{\mathcal{B}}/\zeta)^{1/(H+1/2)}.$$

Proof. Under (i)–(iii), the regularized Sinkhorn value admits a uniform first-order expansion on $\mathcal{B}$ with a remainder lower order than $n^{-1/2}$. The embedded fixed-span support and dual-complexity hypotheses in (iv) supply the same structural comparison needed in the 1D proof model. Applying the abstract horizon law with $a = 1/2$ yields the stated envelope and the induced horizon scale.

Proposition 2.19 (Pointwise fixed-ε inheritance in the 1D proof model)**.** *Assume the hypotheses of proposition 2.8, let $\widehat{P}_n^{\mathrm{iid}}$ be the empirical measure of n i.i.d. samples from $\bar{P}_n$, and fix $\varepsilon > 0$. For the debiased Sinkhorn* *divergence S_ε with squared-Euclidean cost on the common compact support, one has*

$$\mathbb{E}\, S_\varepsilon\left(\widehat{P}_n^{\text{tri}}, \bar{P}_n\right) = O(n^{-1/2}).$$

Proof. Write $V_\varepsilon(\mu,\nu)$ for the regularized OT value so that

$$S_\varepsilon(\mu,\nu) = V_\varepsilon(\mu,\nu) - \tfrac{1}{2}V_\varepsilon(\mu,\mu) - \tfrac{1}{2}V_\varepsilon(\nu,\nu).$$

Because all measures are supported on one compact interval, they lie in a common bounded quadratic-moment class. By Eckstein and Nutz [2022, Thm. 3.7(ii), Ex. 3.4], for fixed ε there exists $L_\varepsilon < \infty$ depending only on that class and on the cost such that

$$|V_\varepsilon(\mu,\nu) - V_\varepsilon(\mu',\nu')| \le L_\varepsilon\big(W_2(\mu,\mu') + W_2(\nu,\nu')\big).$$

Applying this bound to the cross term and both self terms gives

$$|S_\varepsilon(\mu,\nu) - S_\varepsilon(\mu',\nu')| \le 2L_\varepsilon\big(W_2(\mu,\mu') + W_2(\nu,\nu')\big).$$

With $\nu = \nu' = \bar{P}_n$,

$$S_\varepsilon\left(\widehat{P}_n^{\text{tri}}, \bar{P}_n\right) \le S_\varepsilon\left(\widehat{P}_n^{\text{iid}}, \bar{P}_n\right) + 2L_\varepsilon W_2\left(\widehat{P}_n^{\text{tri}}, \widehat{P}_n^{\text{iid}}\right).$$

Taking expectations and using the triangle inequality gives

$$\mathbb{E}\, W_2\left(\widehat{P}_n^{\text{tri}}, \widehat{P}_n^{\text{iid}}\right) \le \mathbb{E}\, W_2\left(\widehat{P}_n^{\text{tri}}, \bar{P}_n\right) + \mathbb{E}\, W_2\left(\widehat{P}_n^{\text{iid}}, \bar{P}_n\right).$$

The first term is $O(n^{-1/2})$ by proposition 2.8. The second is the classical i.i.d. one-dimensional quantile-process rate on the same compact support class [Bahadur, 1966, van der Vaart, 1998], so it is also $O(n^{-1/2})$. Hence

$$\mathbb{E}\, W_2\left(\widehat{P}_n^{\text{tri}}, \widehat{P}_n^{\text{iid}}\right) = O(n^{-1/2}).$$

Finally, under the null comparison with fixed ε and compact support, the i.i.d. debiased Sinkhorn term satisfies

$$\mathbb{E}\, S_\varepsilon\left(\widehat{P}_n^{\text{iid}}, \bar{P}_n\right) = O(n^{-1})$$

by del Barrio et al. [2023, Thm. 5.1, after fixed-ε rescaling]; see also the null-limit law in Goldfeld et al. [2024]. Combining the bounds yields the displayed $O(n^{-1/2})$ rate.

Proposition 2.20 (Bandwise fixed-ε inheritance in the 1D proof model)**.** *Let $\mathcal{B} = [\varepsilon_{\min}, \varepsilon_{\max}] \subset (0,\infty)$ be a compact regularization band. Under the hypotheses of proposition 2.19 on a common compact support class, there exists $C_{\mathcal{B}} < \infty$ such that*

$$\sup_{\varepsilon\in\mathcal{B}} \mathbb{E}\, S_\varepsilon\left(\widehat{P}_n^{\text{tri}}, \bar{P}_n\right) \le C_{\mathcal{B}} n^{-1/2}.$$

Consequently, the induced horizon scale is uniform on $\mathcal{B}$:

$$\sup_{\varepsilon\in\mathcal{B}} n_\varepsilon^*(H) \asymp (C_{\mathcal{B}}/\zeta)^{1/(H+1/2)}.$$

Proof. Repeat the proof of proposition 2.19. The stability constant L_ε from Eckstein and Nutz [2022, Thm. 3.7(ii), Ex. 3.4] is bounded on $\mathcal{B}$ because all measures live in one compact support class and $\varepsilon \ge \varepsilon_{\min} > 0$. The i.i.d. debiased-Sinkhorn benchmark on compact support is also uniform on $\mathcal{B}$ by the fixed-ε asymptotics of Goldfeld et al. [2024] and the compact-support null bounds of del Barrio et al. [2023, Thm. 5.1, after fixed-ε rescaling]. Since the triangular-to-i.i.d. W_2 comparison is independent of ε, the constants in the pointwise proof can be chosen uniformly over $\mathcal{B}$.

Proposition 2.21 (Conditional embedded fixed-ε band inheritance)**.** *Assume the fixed-ε Sinkhorn setting from definition 2.16. Suppose that on a compact band $\mathcal{B} = [\varepsilon_{\min}, \varepsilon_{\max}]$, the centered Sinkhorn scaling map admits a uniform linearization with remainder $o(n^{-1/2})$ and a uniform self-coupling inverse bound. Then there exists $C_{\mathcal{B}} < \infty$ such that*

$$\sup_{\varepsilon\in\mathcal{B}} \mathbb{E}\, S_\varepsilon\left(\widehat{P}_{t,\text{tri}}^{(n)}, \bar{P}_t^{(n)}\right) \le C_{\mathcal{B}} n^{-1/2}.$$

Consequently, on any such band the induced horizon scale satisfies

$$\sup_{\varepsilon\in\mathcal{B}} n_\varepsilon^*(H) \asymp (C_{\mathcal{B}}/\zeta)^{1/(H+1/2)}.$$

Proof. The linearization hypothesis reduces the triangular-array Sinkhorn divergence to its first-order influence term plus an $o(n^{-1/2})$ remainder. The uniform self-coupling inverse bound keeps the influence operator bounded across $\mathcal{B}$, while the triangular-array fluctuation of the empirical measure is root-n on the embedded fixed-span class. The same balance then gives the uniform $n^{-1/2}$ bound and the stated horizon scale.

Conjecture 2.22 (Bandwise fixed-ε Sinkhorn horizon inheritance)**.** *Beyond the one-dimensional fixed-ε band result above, assume the fixed-ε Sinkhorn setting from definition 2.16. Then, on bands where the i.i.d. benchmark stays in the parametric Sinkhorn regime, the triangular-array finite-sample term should satisfy*

$$\mathbb{E}\, S_\varepsilon\left(\widehat{P}_{t,\text{tri}}^{(n)}, \bar{P}_t^{(n)}\right) \le C_\varepsilon n^{-1/2}, \qquad \varepsilon \in \mathcal{B},$$

and the corresponding horizon scale should satisfy

$$n_\varepsilon^*(H) \asymp (C_\varepsilon/\zeta)^{1/(H+1/2)}.$$

More generally, if the i.i.d. benchmark on $\mathcal{B}$ has exponent a_ε, then the operational horizon law should be

$$n_\varepsilon^*(a_\varepsilon, H) \asymp (C_\varepsilon/\zeta)^{1/(a_\varepsilon+H)}.$$

Evidence. The i.i.d. benchmark comes from existing fixed-ε entropic transport results, including fixed-marginal limit theory for the Sinkhorn divergence and intrinsic-complexity adaptation [Goldfeld et al., 2024, Stromme, 2023, Groppe and Hundrieser, 2024]. In the present embedded fixed-span model, support-complexity inheritance is exact and the dual smoothness threshold reduces to $\alpha > k/2$. Calibration then asks whether triangular and i.i.d. rate estimates remain close across the regularization band. Additional one-dimensional smooth fixed-support diagnostics show that triangular and i.i.d. fitted exponents can agree closely on moderate fixed-ε bands, while direct Jacobian diagnostics concentrate the remaining difficulty on uniform control of the self-coupling derivative blocks across nontrivial bands and higher-dimensional embedded classes.

Regular dominated families remain the natural family-level extension class. Local testing geometry and metric comparability suggest the same horizon logic with a modified finite-sample exponent, whereas support-changing and singular families fall outside that regime.

Once a finite-sample regime and a roughness level are specified, the law yields an explicit memory scale and an optimal error scale. Rougher drift and weaker finite-sample rates both force shorter memory and larger residual error.

3 Empirical Signatures

These experiments expose the finite-sample signatures of the temporal-validity horizon, the useful-memory region induced by that horizon, and evidence relevant to the fixed-ε Sinkhorn frontier.

The Gaussian U-curve uses 20 Monte Carlo seeds, the roughness-family sweep uses 12, the fixed-ε Sinkhorn fits log–log slopes over sample sizes $(24, 48, 96, 160)$ on the pairs $(8, 1)$, $(8, 2)$, $(12, 1)$, and $(12, 2)$ using 24 seeds, and the validity-detection experiment uses a piecewise-ramped drift stream with 12 seeds, a fixed operating window, and both ADWIN and Page–Hinkley detectors. The real-stream diagnostic uses the ELEC2 stream and a short anchor block on the marginal distribution of `nswprice`. The empirical goal is qualitative and structural: persistence of shape, ordering, and sign across repeated runs, not precise constant estimation for every curve.

The empirical signatures are fivefold. Finite-memory risk exhibits an interior optimum together with a near-optimal useful-memory band. That optimum moves with temporal roughness. The same shape appears on a real stream. The fixed- ε Sinkhorn frontier receives structured empirical support. Temporal validity can fail before changepoint evidence becomes statistically visible.

3.1 Interior Memory Optimum

The Gaussian horizon sweep isolates the variance–staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the minimum lies at an interior horizon.

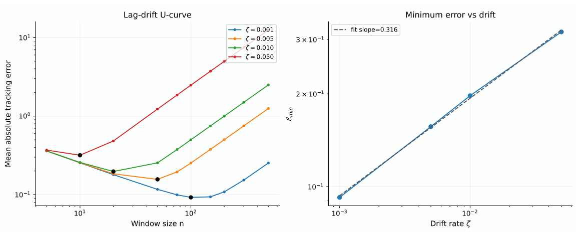

Figure 3: Interior risk optimum over memory length. Tracking error first falls with horizon and then rises once staleness overtakes the remaining variance gain. The optimum remains finite across drift strengths.

This is the basic empirical signature of temporal validity under drift. Under stationarity, more memory would continue to help. Under drift, the same horizon that reduces variance also ages the evidence. Around the minimizing horizon, the curve displays a flat near-optimal region, which is the empirical analogue of the useful-memory region defined from the envelope Φ.

3.2 Roughness-Dependent Horizon Scaling

The roughness-family experiment varies the roughness index directly. For each Hölder-type regime, the surrogate error obeys the same structural decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^{H},$$

and the experiment records the empirically optimal horizon across a sweep of ζ values.

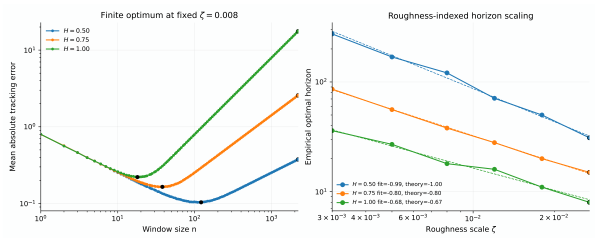

Figure 4: Roughness-dependent horizon scaling. Each path class exhibits a finite optimum, and the location of that optimum changes with H. The horizon is a function of drift regularity, not a fixed window rule.

The empirical optima separate by roughness class, matching the path-dependent horizon logic of corollary 2.5.

3.3 Real-Stream Retention Diagnostic

The same qualitative signature appears on a real non-stationary stream. Using ELEC2, consider the one-dimensional marginal law of `nswprice`. At each time, approximate the present target by a short anchor block of 48 observations and compare it, in W_1, with the empirical law induced by the preceding memory window. Averaging this diagnostic across the stream yields a real-data analogue of the finite-memory U-curve.

On ELEC2, the diagnostic optimum occurs at a finite memory scale, with a short near-optimal band around it. The real stream therefore exhibits the same basic retention pattern as the synthetic horizon experiments: very short windows are too noisy, while very long windows become stale relative to the present block. On a robustness grid with anchor sizes 24, 48, and 96 and three contiguous segments of length 4000, the `nswprice` marginal retains an interior optimum in all nine tested configurations, with edge-to-best error ratios between 1.31 and 1.68. The same interior pattern appears on the `nswdemand` marginal in all nine tested configurations, although the optimizing scale moves more substantially across segments and anchor lengths; see Table 7.

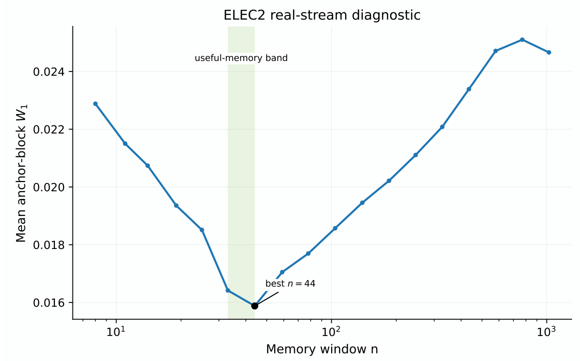

Figure 5: ELEC2 real-stream diagnostic on the `nswprice` marginal. The mean anchor-block W_1 error exhibits a finite interior optimum and a near-optimal useful-memory band. The figure is diagnostic: it asks whether the temporal-validity signature appears on a real stream.

3.4 Moderate-Band Sinkhorn Evidence

The fixed-ε Sinkhorn experiments address the bandwise question: when a practical geometry replaces raw W_2, does the finite-sample term remain stable enough for the horizon law to stay informative in an embedded operational regime? The pointwise and bandwise one-dimensional fixed-support statements appear in propositions 2.19 and 2.20; the calibrated embedded fixed-span model extends the same question to a broader low-dimensional setting.

The fixed-ε Sinkhorn geometry on embedded low-intrinsic-dimensional support shows that the corresponding finite-sample term remains stable across the tested regularization values and that the triangular slope stays close to the i.i.d. benchmark, with no separate exponent emerging. Combined with the structural ingredients isolated in proposition 2.17, this identifies a stable embedded operating regime. In the calibrated fixed-span model with $\alpha = 2$ and span 0.25, the maximal observed stable bands are $\varepsilon_{\max} = 0.5$ for $(d, k) = (8, 1)$, $(8, 2)$, and $(12, 2)$, and $\varepsilon_{\max} = 0.2$ for $(12, 1)$.

The same synthetic family also separates two mechanisms. Raw W_2 comparison between the triangular and i.i.d. empirical measures slows once intrinsic dimension increases, so the embedded fixed-ε statement does not follow from the available W_2 transfer route. Direct Jacobian probes of the centered Sinkhorn scaling map instead isolate a moderate-band regime: on $\varepsilon \in [0.2, 0.8]$, the cross-coupling block stays contractive, while the self-coupling blocks are the main source of instability but appear numerically to plateau with sample size.

The calibrated fixed-span model shows that the tested grid has a stable core and a small mixed boundary region. Calibration details and supporting tables are kept in appendix B. The strongest support appears on moderate bands bounded away from zero.

The certified-band report on the same calibrated grid is positive on the same stable pairs: it returns maximal certified bands of at least 0.5 for $(d, k) = (8, 1)$, $(8, 2)$, and $(12, 2)$, and 0.2 for $(12, 1)$. This does not prove the embedded theorem, but it identifies a concrete moderate-band regime in which the conditional stability assumptions are simultaneously satisfied by the current certificate.

Ambient / intrinsic	α	Max certified ε
$(8, 1)$	2.0	0.5
$(8, 2)$	2.0	0.5
$(12, 1)$	2.0	0.2
$(12, 2)$	2.0	0.5

Table 1: Calibrated moderate-band Sinkhorn certificates on the embedded fixed-span grid. The table records the largest regularization value for which the current certificate returns a certified band on the tested synthetic frontier.

To probe whether the frontier is controlled more by local intrinsic geometry than by the ambient coordinate count, we estimate the effective intrinsic dimension with TwoNN on the same synthetic frontier family. On the nine-pair, six-ε grid studied here, the resulting $\hat{k}$ predicts the stable-band cutoff better than ambient dimension and improves the binary cut at $\varepsilon = 0.2$. This is consistent with geometry-aware calibration: the relevant control variable is the estimated local geometry, not the ambient coordinate count. The estimator therefore enters here as a regime-selection diagnostic for the band analysis.

3.5 Validity Precedes Detection

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over time. The horizon n_t^* contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\mathrm{val-det}} := t_{\mathrm{detect}} - t_{\mathrm{valid}},$$

where t_{valid} is the first time the operating horizon persistently exceeds the temporal-validity horizon, and t_{detect} is the first detector alarm that follows.

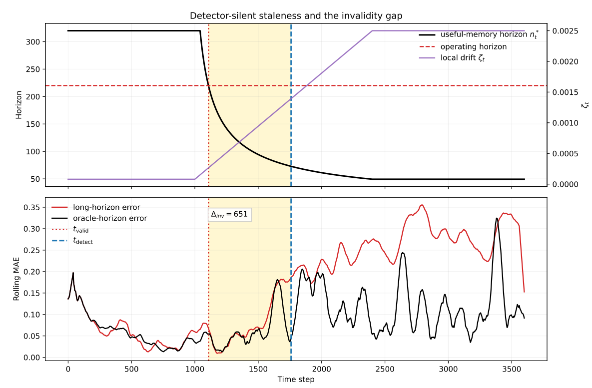

Figure 6: Temporal validity can fail before detector alarm. As drift intensifies, the temporal-validity horizon contracts until the operating horizon becomes stale at t_{valid}; the detector responds only later at t_{detect}, leaving a strictly positive validity-detection lag. The lower panel shows excess tracking error accumulating before the alarm appears.

Across the reported runs, the measured lag remains positive. Retained evidence can already be invalid for the present while detector statistics remain below alarm threshold.

The positive lag is not a knife-edge feature of one tuning choice. Across the tested ADWIN thresholds, mean detection time remains strictly later than mean validity-expiry time, and the same qualitative lag persists under a different detector family as well. On the raw-observation input, the full tested grid of ADWIN and Page–Hinkley detectors, operating windows 160, 220, and 300, and the reported threshold sweep keeps positive-gap rate equal to one together with unit detection rate. Page–Hinkley reacts earlier on this drift family, with mean gaps between about 205 and 345, while ADWIN yields mean gaps between about 429 and 620. Residualized detector inputs are not uniformly better on the same stream: ADWIN on absolute residuals fails to trigger on the tested grid, whereas Page–Hinkley still detects but with lower detection rates and substantially larger delays. The empirical claim is robustness of the sign of $\Delta_{\text{val}-\text{det}}$ on the tested observation-based detector/tuning family, not a detector-uniform delay law; see Table 8.

3.6 Empirical Signatures

The experiments support a structural U-curve with a near-optimal useful-memory region, roughness-dependent horizon scaling, the same retention pattern on the ELEC2 stream, structured support for the fixed-ε Sinkhorn frontier, and a measurable lag between validity loss and detector alarm.

4 Inferable Horizons

The horizon law is structural, but its operational use requires more than the existence of an optimizer. A data-driven memory rule needs estimators of the drift-geometry parameters (H, ζ), together with a quantitative criterion that compares inferential resolution to the curvature of the useful-memory region. The resulting bridge turns temporal validity into an inferable object.

Write $\alpha = \log \zeta$, and let m denote the per-lag sample size used to estimate lagged transport discrepancies. For a fixed lag design $1, \ldots, L$, define the lag-energy information scale

$$\mathcal{I}_{m,L}(H) := m \sum_{j=1}^{L} j^{2H}.$$

Assumption 4.1 (Lag-geometry inference input)**.** For the lag model $D_j = \zeta j^H$, suppose there are estimators $(\widehat{\alpha}_{m,L}, \widehat{H}_{m,L})$ such that, on compact parameter sets,

$$\sqrt{\mathcal{I}_{m,L}(H)} \begin{pmatrix} \widehat{\alpha}_{m,L} - \alpha \\ \widehat{H}_{m,L} - H \end{pmatrix} \Rightarrow \mathcal{N}(\mathbf{0}, \Sigma_{L,H,\zeta}).$$

The power-law lag-geometry model of Parreira [2026b] provides one concrete source of this assumption.

The continuous optimizer of the horizon envelope can be written as the smooth map

$$n_\star(\alpha, H) = \left(\frac{aC_K}{HC_S e^\alpha} \right)^{1/(a+H)},$$

$$g(\alpha, H) := \log n_\star(\alpha, H).$$

Its gradient is

$$\nabla g(\alpha, H) = \begin{pmatrix} -(a+H)^{-1} \\ -\left[H(a+H)\right]^{-1} - g(\alpha, H)(a+H)^{-1} \end{pmatrix}.$$

Theorem 4.2 (Plug-in Horizon Central Limit Law)**.** *Under Assumption 4.1,*

$$\sqrt{\mathcal{I}_{m,L}(H)} \big(\log \widehat{n}_\star - \log n_\star\big) \Rightarrow \mathcal{N}(0, \tau^2),$$

where $\widehat{n}_\star = n_\star(\widehat{\alpha}_{m,L}, \widehat{H}_{m,L})$ and

$$\tau^2 = \nabla g(\alpha, H)^\top \Sigma_{L,H,\zeta} \nabla g(\alpha, H).$$

Proof. The map g is smooth on any compact set bounded away from $H = 0$ and $\zeta = 0$. Apply the multivariate delta method to $(\widehat{\alpha}_{m,L}, \widehat{H}_{m,L})$ under Assumption 4.1.

The useful-memory region is the set

$$\mathcal{U}_\delta = \{n : \Phi(n) \le (1+\delta)\Phi(n_\star)\}.$$

Let $(x_-(\delta; a, H), x_+(\delta; a, H))$ be the normalized endpoints from Proposition 2.6, and define the log-radius

$$r_\delta(a, H) := \min\big\{-\log x_-(\delta; a, H),\, \log x_+(\delta; a, H)\big\}.$$

This quantity measures the operational tolerance of the horizon law: it is the largest symmetric log-error around $n_\star$ that stays inside the useful-memory region.

Theorem 4.3 (Operational Identifiability of Useful Memory)**.** *Under Assumption 4.1,*

$$\mathbb{P}\big(\widehat{n}_\star \in \mathcal{U}_\delta\big) = 2\Phi_{\mathcal{N}}\left(\frac{\sqrt{\mathcal{I}_{m,L}(H)}\, r_\delta(a, H)}{\tau} \right) - 1 + o(1),$$

where $\Phi_{\mathcal{N}}$ is the standard normal CDF. In particular,

$$\frac{\sqrt{\mathcal{I}_{m,L}(H)}\, r_\delta(a, H)}{\tau(H, \zeta)} \to \infty \quad \Longrightarrow \quad \mathbb{P}\big(\widehat{n}_\star \in \mathcal{U}_\delta\big) \to 1.$$

Proof. By Proposition 2.6, $\widehat{n}_\star \in \mathcal{U}_\delta$ if and only if $|\log \widehat{n}_\star - \log n_\star| \le r_\delta(a, H)$. Standardize this event with Theorem 4.2 and use the symmetry of the Gaussian limit.

The quantity

$$\mathfrak{S}_{\delta,m,L} := \frac{\sqrt{\mathcal{I}_{m,L}(H)}\, r_\delta(a, H)}{\tau(H, \zeta)}$$

is therefore the operational identifiability score of the horizon. It compares lag-geometry information to the curvature-imposed precision requirement of the useful-memory region.

Proposition 4.4 (Deterministic stability of the horizon map)**.** *Let $K \subset \mathbb{R} \times (0, \infty)$ be compact. Then there exists a finite constant L_K such that, for all $(\alpha_1, H_1), (\alpha_2, H_2) \in K$,*

$$|g(\alpha_1, H_1) - g(\alpha_2, H_2)| \le L_K\big(|\alpha_1 - \alpha_2| + |H_1 - H_2|\big).$$

Consequently, if (α, H) and $(\widehat{\alpha}_{m,L}, \widehat{H}_{m,L})$ lie in K and

$$|\widehat{\alpha}_{m,L} - \alpha| + |\widehat{H}_{m,L} - H| \le \frac{r_\delta(a, H)}{L_K},$$

then $\widehat{n}_\star \in \mathcal{U}_\delta$.

Table 2: Representative regimes across the inferable-horizon transition.

L	m	H	δ	$\mathfrak{S}$	Empirical	Gaussian
5	20	0.3	0.050	0.501	0.411	0.384
8	10	0.9	0.020	1.001	0.633	0.683
3	200	0.3	0.010	1.996	0.956	0.954
8	100	0.3	0.005	3.001	0.996	0.997

Proof. The gradient of g is continuous on K, so $\|\nabla g\|_1$ is bounded there. Apply the mean-value theorem to obtain the Lipschitz bound. The inclusion claim follows from $|\log \widehat{n}_\star - \log n_\star| \le r_\delta(a,H)$ together with the characterization of $\mathcal{U}_\delta$ from Proposition 2.6.

The informative regime lies across the transition from low to moderate identifiability. Table 2 records representative points across that transition.

On the tested grid, the plug-in central limit law is also well calibrated at the scale level: the empirical standard deviation of $\log \widehat{n}_\star$ tracks the delta-method prediction closely, and the nominal 95% Gaussian interval coverage stays near its target across the reported regimes. The same simulation suite gives a direct calibration check for the regret expansion.

Proposition 4.5 (Deterministic and asymptotic plug-in horizon regret)**.** *Let $u = \log(\widehat{n}_\star/n_\star)$. Then, as $u \to 0$,*

$$\frac{\Phi(\widehat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)} = \Psi(e^u) - 1 = \frac{He^{-au} + ae^{Hu}}{a+H} - 1.$$

Consequently, for every $r > 0$,

$$|u| \le r \quad \Longrightarrow \quad \frac{\Phi(\widehat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)} \le B_r(a,H),$$

where

$$B_r(a,H) := \max\{\Psi(e^r) - 1,\ \Psi(e^{-r}) - 1\}.$$

Moreover,

$$\frac{\Phi(\widehat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)} = \frac{aH}{2}u^2 + O(u^3).$$

Consequently,

$$\mathbb{E}\left[\frac{\Phi(\widehat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)}\right] = \frac{aH}{2}\frac{\tau^2}{\mathcal{I}_{m,L}(H)} + o\big(\mathcal{I}_{m,L}(H)^{-1}\big).$$

Proof. By Proposition 2.6,

$$\frac{\Phi(n_\star e^u)}{\Phi(n_\star)} = \Psi(e^u) = \frac{He^{-au} + ae^{Hu}}{a+H}.$$

The deterministic bound follows by maximizing this exact expression over $u \in [-r, r]$. Differentiate at $u = 0$. The first derivative vanishes and the second derivative equals aH, giving the quadratic expansion. The expectation bound then follows from Theorem 4.2 and the boundedness of third moments on compact parameter sets.

This section separates two questions that need not agree. A temporal-validity horizon may exist structurally without being operationally identifiable from the available lag geometry. The gap between existence and operational inferability is governed by the score $\mathfrak{S}_{\delta,m,L}$.

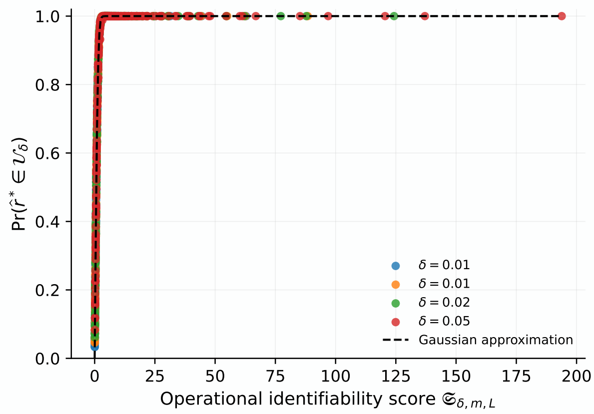

Figure 7: Operational identifiability of useful memory. Across the tested grid, useful-region hit rates collapse against the score $\mathfrak{S}_{\delta,m,L} = \sqrt{\mathcal{I}_{m,L}(H)}r_\delta(a,H)/\tau(H,\zeta)$, matching the Gaussian approximation from Theorem 4.3.

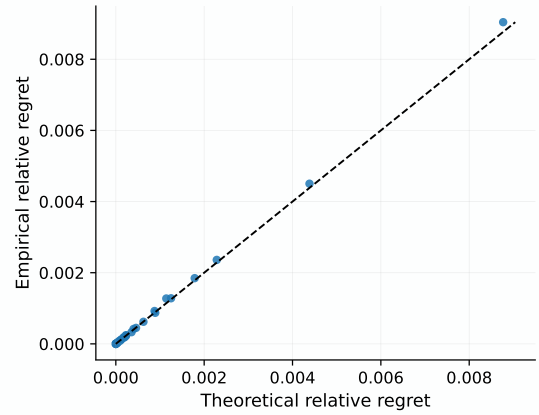

Figure 8: Quadratic operational regret. Across the tested grid, the empirical relative regret of the plug-in horizon aligns with the quadratic law from Proposition 4.5, showing that horizon mislocation costs are second order in log-scale error in the theorem regime.

5 Implications for Online Adaptation

An online memory rule needs three ingredients: a finite-sample regime, a local drift geometry, and a criterion that relates estimation uncertainty to the curvature of the useful-memory region. The horizon law supplies the target once the regime parameter a and the local roughness pair (H,ζ) are known. The inferable-horizon results identify the batch requirement: lag-geometry information must be strong enough to place the plug-in horizon inside the useful-memory region with high probability. The remaining online question is how to maintain that requirement under streaming noise and local nonstationarity.

In operational transport geometries, local support structure enters before roughness estimation. The embedded fixed-ε theory shows that intrinsic complexity controls which finite-sample regime is available through the threshold $\alpha > k/2$. A local intrinsic-dimension diagnostic such as TwoNN therefore acts as a regime-selection statistic: it does not estimate the horizon directly, but it selects the finite-sample law to which the horizon map should be attached.

The roughness side is a separate statistical task. In deterministic Hölder paths, H and ζ govern how quickly staleness accumulates with lag. A single short log–log fit is too unstable to support online control reliably: local motion, finite-sample noise, and short-lag discreteness all distort the envelope. The natural online object is therefore a stabilized lag-geometry estimate built from aggregated recent lags. The companion lag-geometry paper [Parreira, 2026b] gives the batch inferential law for that step; the open problem here is to obtain a rolling or recursively updated version with local stationarity guarantees.

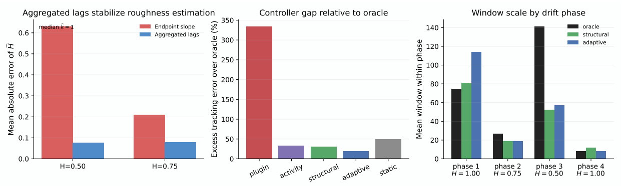

Figure 9: Embedded geometry and online adaptation. Left: aggregated lag statistics estimate H far more accurately than endpoint slopes in the tested low-intrinsic-dimensional regime. Middle: excess tracking error over the oracle controller, showing that the structural controller improves decisively on the direct plug-in rule and on the activity and static baselines, while the broader adaptive hybrid remains strongest. Right: oracle, structural, adaptive, and phase-averaged windows across the four drift phases of the default profile.

The comparison in Figure 9 separates two controller architectures. A direct plug-in rule based on a local slope and intercept is too unstable to be useful. A structural controller first selects the finite-sample regime, then estimates roughness from aggregated lag discrepancies, and finally maps those quantities into an admissible useful-memory band. On the synthetic phase profile, that controller improves substantially on the naive plug-in rule and on a pure activity baseline, although the broader validation-based hybrid remains more accurate.

On the tested default and rough phase profiles with two seeds and a bounded window-jump constraint, the structural controller improves on the direct plug-in rule in all runs and beats the activity baseline in three of the four runs. The validation-based hybrid remains strongest on the default profile, but it is not uniformly best on the rough profile. The tested profiles therefore support a separation between unstable direct plug-in control and the more stable band-constrained structural rule, without establishing theorem-level dominance for any single online controller; see Table 9.

On any compact parameter range with H bounded away from zero and ζ bounded above and below, the map

$$(H, \zeta) \mapsto n_\star(H, \zeta) = \left(\frac{aC_K}{HC_S\zeta} \right)^{1/(a+H)}$$

is Lipschitz in $(H, \log \zeta)$ after taking logarithms. The inferable-horizon section sharpens this qualitative statement: online adaptation is viable only when the rolling lag-geometry estimate is precise relative to the log-radius of the useful-memory region. The remaining difficulty is the scale variable: a local activity statistic is good enough to produce a useful controller, but not yet stable enough to match the stronger validation-based hybrid.

The online problem is therefore a two-error problem. One term comes from statistical uncertainty in the estimated lag geometry; the other comes from the parameter drift accumulated within the estimation window itself. The horizon law supplies the target for adaptation, and the inferable-horizon results supply the batch bridge from lag geometry to memory choice. A time-uniform guarantee that combines those two errors and keeps a data-driven memory rule inside the useful-memory region remains open.

6 Related Work

Nonstationary optimization and dynamic regret. Dynamic-regret and variation-budget analyses show that under drift, forgetting scales and window lengths must adapt to how quickly the target moves [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. That literature isolates the control importance of retention scale. The horizon law gives the corresponding statistical law for distribution tracking: finite memory has a temporal-validity horizon determined jointly by a finite-sample exponent and a drift roughness exponent.

This distinction is especially sharp relative to adaptive update policies such as Mazzetto and Upfal [2023]. Those works optimize a decision or intervention rule under evolving data. The present object is earlier in the pipeline: it asks how long retained evidence continues to represent the present distribution before age becomes the dominant source of tracking error.

Adaptive windowing, drift detection, and update policies. Adaptive-window and concept-drift methods study when accumulated evidence is sufficient to react [Bifet and Gavaldà, 2007, Gama et al., 2004, 2014]. Hinder et al. [2023] emphasize that the window length itself is part of the problem. Resource-aware update scheduling under drift addresses a related decision problem in which update cost must be balanced against accumulated loss [Piaseczny et al., 2025], and finite-time analyses of sliding-window or discounted adaptation under switching environments make the same design variable explicit from the regret side [Garivier and Moulines, 2011]. Streaming OT and MMD detectors contribute the alarm-time perspective, emphasizing delay, power, and memory usage [Shvetsov et al., 2020, Kalinke et al., 2025]. These works provide the reaction timescale. The horizon law adds the validity timescale: retained evidence can cease to represent the present before change statistics cross an alarm threshold.

Adaptive filtering and locally stationary smoothing. Classical adaptive filtering and state-space tracking tune forgetting factors, gain schedules, or process-noise levels to balance smoothing against responsiveness [Kalman, 1960, Jazwinski, 1970]. Locally stationary smoothing and bandwidth selection solve the analogous bias–variance problem for evolving regression or spectral targets [Dahlhaus, 1997, Vogt, 2012]. Recent Wasserstein-based results for locally stationary functional time series move that same local windowing logic directly into a transport metric [Tinio et al., 2025]. These literatures supply mechanisms for adaptation. The horizon law provides the distributional law that such mechanisms must respect when the retained object is an empirical distribution rather than a latent state or a local regression surface.

Lag-geometry inference under noisy transport observations is complementary to that structural law. Parreira [2026b] study power-law lag geometry through a parameter-coupled heteroscedastic regression, deriving an information law for estimating (H, ζ) and testing the adequacy of the lag model itself. That inferential layer supplies one route for estimating the roughness inputs that enter the horizon law developed here.

Distribution tracking under transport geometry. Wasserstein distance and Sinkhorn-type estimators provide the finite-sample geometries that enter the horizon law. For empirical Wasserstein distances, Fournier and Guillin [2015], Boissard and Le Gouic [2014], and Weed and Bach [2019] show that the sampling exponent depends on geometric complexity even before drift enters. The horizon law takes that finite-sample exponent as one input and composes it with a staleness exponent induced by the drift path. In this sense, the transport literature provides the sampling side of the law, while the horizon law provides its temporal counterpart.

Entropic and smoothed transport as operational geometries. Regularized transport changes both statistical stability and computational behavior [Genevay et al., 2019, Mena and Niles-Weed, 2019, del Barrio et al., 2023, Rigollet and Stromme, 2025, Stromme, 2023, Groppe and Hundrieser, 2024, Eckstein and Nutz, 2022, Ghosal et al., 2022]. Fixed-ε Sinkhorn theory provides i.i.d. asymptotics, differentiability, null limit laws, and stability constants [Goldfeld et al., 2024]. Those results are the finite-sample anchor for the operational horizon law developed here. Recent linear-Gaussian entropic- OT analyses make the recursive structure of regularized transport explicit [Akyildiz et al., 2024], suggesting a constructive intermediate regime between fixed-ε i.i.d. asymptotics and fully drifted triangular-array tracking.

Smooth-functional and delta-method arguments provide the multivariate transfer mechanism behind this regime. Hadamard differentiability is the natural regularity class for lifting triangular-array fluctuations through a transport functional, and the fixed-ε Sinkhorn map sits in that category after gauge fixing and uniform stability control [Goldfeld et al., 2024]. That route is the right multivariate complement to the one-dimensional quantile-process proof model.

Intrinsic geometry and operational calibration. Geometry-aware calibration of regularization scales points toward intrinsic dimension as the relevant complexity variable in regularized transport [Genans and Wintenberger, 2026]. Local intrinsic-dimension diagnostics such as TwoNN provide a direct statistic for effective support geometry [Facco et al., 2017]. That perspective enters through the operational Sinkhorn frontier: intrinsic geometry helps identify which finite-sample regime is available, and therefore which horizon law is relevant.

Low-dimensional surrogate geometries. Sliced, max-sliced, subspace-robust, and projection-robust Wasserstein constructions replace ambient transport by one-dimensional or low-dimensional comparisons [Kolouri et al., 2018, Deshpande et al., 2019, Paty and Cuturi, 2019, Lin et al., 2020, Nadjahi et al., 2020, Nietert et al., 2022]. These examples show concretely that geometry choice can change the finite-sample exponent. They therefore fit naturally into the horizon framework as alternative sampling geometries whose drift-induced useful-memory scales need not match raw Wasserstein behavior.

Wasserstein nonstationarity and Age of Information. Wasserstein nonstationarity measures already appear in online optimization and distributionally robust optimization under drift [Jiang et al., 2020, Keehan et al., 2025, Amini and Vinas, 2026]. The Age-of-Information literature treats age as a resource that must be balanced against communication or update costs [Kaul et al., 2012, Yates et al., 2021]. Together these viewpoints make age and nonstationarity into operational variables. The horizon law gives them a distributional statistical interpretation: age is costly because it accumulates geometric mismatch with the present, and the roughness exponent determines how fast that cost grows.

Keehan et al. [2025] are particularly close in spirit: they derive concentration bounds for weighted empirical laws under sequential Wasserstein drift and optimize weights for distributionally robust decisions. That question begins once a decision rule is fixed. The temporal-validity horizon addresses the preceding statistical question of how long evidence should be retained before staleness dominates the finite-sample gain. In this sense, their weighting problem and the present horizon problem are complementary rather than competing.

Weighted empirical-process and weighted-Wasserstein precursors are also relevant for the non-uniform-memory line. In particular, Mason [2016] studies weighted approximations to empirical Wasserstein fluctuations, providing a technical ancestor for the weighted-memory summaries $n_{\text{eff}}(w)$ and $M_H(w)$ used here.

Across these neighboring literatures, the shared theme is that nonstationarity makes retention scale matter. The horizon law identifies the statistical object that organizes that dependence for distribution tracking: a temporal-validity horizon, together with a useful-memory region and a distinct detectability timescale.

7 Limitations and Open Problems

The central structural object is the temporal-validity horizon for finite-memory tracking under drift, and the closed operational corollary is the induced useful-memory region. The theorem-level results are the general horizon law, the useful-memory region proposition, the uniform-window staleness bound, the one-dimensional root-n proof model, the structural Gaussian lower bound at the exponent level, the Gaussian location minimax benchmark, and the plug-in inferable-horizon laws for coverage and regret under lag-geometry input. The remaining claims are narrower.

Finite-sample scope. The theory is modular in the finite-sample exponent a, but the fully proved finite-sample instantiation established here is the one-dimensional root-n regime under bounded support and fixed span. The same balancing logic extends to other geometries only once an appropriate finite-sample theorem is supplied.

Operational scope. For fixed-ε Sinkhorn, the structural ingredients of the embedded fixed-span model are proved, and the compact-support one-dimensional pointwise and bandwise inheritance theorems are explicit. The remaining operational step is the embedded low-dimensional theorem that connects the triangular-array finite-sample term to the i.i.d. benchmark uniformly across a nontrivial regularization band. The numerical evidence is strongest on moderate compact bands bounded away from zero, where direct Jacobian probes remain stable.

Lower-bound scope. The structural lower bound is subclass based. The Gaussian construction is enough to show that, in the root-n regime, the horizon is structural rather than an artifact of a particular estimator. Class-tightness and constant- sharpness for broader deterministic Hölder path classes remain outside the present closure.

Sharp-constant and lower-theory scope. The exact Gaussian lower-bound constants are explicit within the proof scheme, but they are not class-tight across broader path classes. Extending the lower theory beyond the root-n regime, to other finite-sample exponents $a \neq 1/2$, and to other parametric deformations such as Gaussian scale remains open.

Family-level scope. Finite families of one-dimensional projections, including coordinate-averaged product Wasserstein constructions, are covered theoremically under uniform projected control. Richer dominated families remain the natural extension class when local testing geometry and metric geometry stay comparable. Support-changing and singular families remain obstruction classes.

Weights and memory shape. Uniform windows are the main case. The open question is whether broader non-uniform weights can enlarge the useful-memory region or improve constants without changing the exponent across richer path classes. The minimal tapered-weight diagnostic is consistent with a modest constant-level gain on the default synthetic stream, but it is still only a single-stream signal and does not yet support a general non-uniform-memory claim. The theorem-oriented weighted summaries in the main text isolate the relevant quantities as $n_{\text{eff}}(w)$ and $M_H(w)$; the remaining question is whether those summaries support a robust constant improvement beyond the uniform window on more than one drift profile.

Online roughness estimation. The inferable-horizon section gives a batch plug-in law that propagates uncertainty from lag-geometry estimation into the selected horizon. What remains open is a fully online procedure for estimating H and ζ and for keeping the memory inside the useful region under misspecification.

Empirical scope. The experiments are structurally motivated. Synthetic designs make the horizon law, useful-memory region, and validity-detection lag visible under controlled drift, while the ELEC2 diagnostic shows that the same retention signature appears on a real stream. Their role is to expose the object, not to demonstrate competitive performance on a benchmark suite.

Detectability versus validity. Validity loss before detector alarm is an empirical consequence of the theory, not the primary theorem object. Temporal validity can fail before detector alarms appear, but it does not prove a universal detection-delay theorem across detector classes.

Open problems. Five directions organize the remaining frontier: operational theorem closure, sharp lower theory, memory design, online roughness estimation, and online control together with validity-detection theory.

Problem 1: operational theorem closure for embedded Sinkhorn and richer multivariate transfer. Embedded Sinkhorn has a conditional moderate-band theorem, a certified band report, and a self-coupling stability certificate. The remaining problem is to prove the self-coupling stability bound in the embedded fixed-span model, and to lift the current finite-family projection and smooth-transfer results to a broader transport-functional theorem with a verified Jacobian remainder bound.

Problem 2: sharp lower theory beyond the benchmark family. The exact Gaussian constants are available within the proof scheme, but the class-tight lower-bound problem remains open for broader deterministic path classes and for alternative parametric deformations. The open problem is to extend the lower theory beyond the root-n regime, beyond the location benchmark, and beyond the available class-level sharp constants.

Problem 3: useful-memory design beyond uniform windows. The analysis is built around uniform windows, but the weighted-memory theorem shows that uniform weights maximize effective sample size and linear ramps recover exact lag. The open problem is to determine when non-uniform weights can improve constants or enlarge the useful-memory region on richer path classes, and whether the summaries $n_{\text{eff}}(w)$ and $M_H(w)$ support a robust optimization law of the form

$$\min_{w:\,\|w\|_1=1} \left[C_K n_{\text{eff}}(w)^{-a} + C_S \zeta M_H(w) \right].$$

This is the memory-design problem that links static window design to fully adaptive control.

Problem 4: online estimation of roughness and scale. A batch plug-in horizon law is available once a lag-geometry CLT is given. The remaining step is to derive convergence rates for online estimators of H and ζ when transport discrepancies are themselves observed with finite-sample error. A structural controller, an intrinsic-dimension statistic, and an aggregated-lag roughness estimate are available. A concrete target is the log–log estimator

$$\widehat{H}_L \in \arg\min_H \sum_{j=1}^{L} \bigl(\log \widehat{D}_j - \log \widehat{\zeta} - H \log j\bigr)^2,$$

where $\widehat{D}_j$ estimates the lag-j transport discrepancy. The open problem is to derive the rate of $\widehat{H}_L$ as a function of L, the finite-sample exponent a, and the lag-noise level, and then to propagate that rate through the horizon map to obtain a high-probability statement of the form $\mathbb{P}(\widehat{n}_t \in \mathcal{U}_\delta) \geq 1 - \eta_{L,n}$.

Problem 5: online control and validity-detection theory. The positive validity-detection lag suggests that a memory controller can, in principle, react before standard alarms fire. The open problem is to prove an excess-risk guarantee of the form

$$\mathbb{E}\,\Phi(\widehat{n}_t) \leq (1 + \delta_t)\Phi(n_t^*)$$

for an adaptive window rule, and to derive detector-specific lower or upper laws that quantify when validity must fail before statistical detection. Under stylized ramp drift, a natural target is a detector-specific law for the expected lag $\Delta_{\mathrm{val-det}}$, for example a scaling relation of the form $\mathbb{E}\,\Delta_{\mathrm{val-det}} \asymp (\zeta/\sigma)^{-2/(2H+1)}$ up to detector-dependent constants.

8 Conclusion

Under drift, memory is never purely beneficial. It reduces finite-sample noise, but it also ages. The resulting trade-off makes finite memory a temporal-validity object with a finite horizon.

That idea is formalized through the horizon law

$$\mathbb{E}\,d\Bigl(\widehat{P}_t^{(n)}, P_t\Bigr) \leq C_K n^{-a} + C_S \zeta n^H,$$

whose optimal scale is

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

Finite-memory tracking under drift is governed by this balance between finite-sample accuracy and drift-induced staleness, and the same balance induces a useful-memory region around the optimal scale.

A tractable one-dimensional proof model supplies a rigorous root-n finite-sample regime, and a structural Gaussian lower-bound construction shows that the corresponding horizon is not an artifact of one estimator. The same object yields a Gaussian location minimax benchmark on deterministic Hölder paths. In fixed-ε Sinkhorn geometry, the compact-support one-dimensional pointwise and bandwise inheritance theorems show that the same horizon logic remains visible in an operational transport geometry.

The same decomposition extends to non-uniform lag weights through effective sample size and weighted lag moment summaries. The theorem-level question is whether that weighted horizon law improves constants on richer drift classes or only reshapes the same balance.

The finite-memory U-curve, roughness-dependent horizon scaling, a useful-memory region around the optimal scale, and a strictly positive lag between validity loss and detector alarm all fit the same horizon law. The resulting message is both theoretical and operational: retained evidence has a path-dependent horizon, useful memory occupies a finite near-optimal region, and statistical detectability need not coincide with temporal validity.

The inferable-horizon layer adds a second distinction. A temporal-validity horizon can exist structurally and yet remain operationally non-identifiable from the available lag geometry. The plug-in horizon law, the useful-region coverage criterion, and the quadratic regret expansion identify the information scale on which a data-driven memory choice becomes reliable.

Temporal validity of retained evidence is therefore a first-class statistical object under drift. The horizon law identifies its governing scale, the useful-memory region records its operational range, and the detectability lag shows that validity can fail before alarms appear. The inferable-horizon bridge adds a fourth statement: operational memory choice is controlled jointly by lag- geometry information and by the curvature of the useful-memory region.

The five open problems in section 7 mark the precise frontier of this program. The first two close the structural theorem line, the third isolates the design limits of memory itself, and the last two turn the horizon law into a fully online object with estimation and control guarantees. Together they define a coherent next stage for temporal validity under drift.

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A Proof Details

This appendix gives the detailed derivations for structural results whose full calculations would interrupt the main line.

A.1 Gaussian lower-bound constants

Let

$$S_h := \sum_{r=1}^{h} r^{2H} \qquad \text{and} \qquad G_h := \sum_{r=1}^{h} \big(h^H - (h-r)^H\big)^2.$$

For the Gaussian two-point model with equal priors and common covariance $\sigma^2 I_h$, the Bayes classification error between means $\pm m$ is $\Phi(-\|m\|/\sigma)$. Therefore the ramp and endpoint-minimal payoffs are exactly

$$L_h^{\mathrm{ramp}}(\sigma,\zeta) = \zeta h^H \Phi\Big(-\frac{\zeta}{\sigma}\sqrt{S_h}\Big)$$

and

$$L_h^{\min}(\sigma,\zeta) = \zeta h^H \Phi\Big(-\frac{\zeta}{\sigma}\sqrt{G_h}\Big).$$

Proof of proposition 2.12. Fix $H \in (0,1]$ and write $p_H = 2H/(2H+1)$.

For the ramp profile,

$$S_h \sim \frac{h^{2H+1}}{2H+1}.$$

Set

$$h = A\left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)}$$

with $A > 0$ fixed. Then

$$\frac{\zeta}{\sigma}\sqrt{S_h} \to \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

At the same time,

$$\zeta h^H = A^H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Hence

$$\frac{L_h^{\mathrm{ramp}}(\sigma,\zeta)}{\sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}} \to A^H \Phi\Big(-\frac{A^{H+1/2}}{\sqrt{2H+1}}\Big).$$

Define

$$x = \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

Then

$$A^H = (2H+1)^{H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$(2H+1)^{H/(2H+1)} x^{p_H} \Phi(-x).$$

Its logarithmic derivative is

$$\frac{p_H}{x} - \frac{\varphi(x)}{\Phi(-x)}.$$

Since the inverse Mills ratio $\varphi(x)/\Phi(-x)$ is strictly increasing on $(0,\infty)$, this objective has a unique maximizer $x_H > 0$, characterized by

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

This proves

$$\sup_{h\ge 1} L_h^{\mathrm{ramp}}(\sigma,\zeta) \sim C_H^{\mathrm{ramp}}\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\mathrm{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

For the endpoint-minimal profile, the Hölder constraint and endpoint condition imply

$$g_r \ge g_h - (h-r)^H = h^H - (h-r)^H$$

for every feasible endpoint-saturating profile g. Therefore the profile

$$g_r^{\min} = h^H - (h-r)^H$$

minimizes testing energy pointwise and hence minimizes $\sum_{r=1}^h g_r^2$. Its rescaled energy satisfies the Riemann-sum limit. Thus

$$h^{-(2H+1)} G_h \to I_H,$$

where

$$I_H = \int_0^1 \big(1-(1-u)^H\big)^2\, du = \frac{2H^2}{(H+1)(2H+1)}.$$

With the same scaling

$$h = A\left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)},$$

one has

$$\frac{\zeta}{\sigma}\sqrt{G_h} \to A^{H+1/2}\sqrt{I_H}.$$

Setting $x = A^{H+1/2}\sqrt{I_H}$ gives

$$A^H = I_H^{-H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$I_H^{-H/(2H+1)} x^{p_H} \Phi(-x).$$

The same maximizer x_H therefore yields

$$\sup_{h\ge 1} L_h^{\min}(\sigma,\zeta) \sim C_H^{\min}\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Finally,

$$\frac{C_H^{\min}}{C_H^{\mathrm{ramp}}} = \left(\frac{I_H^{-1}}{2H+1}\right)^{H/(2H+1)} = \left(\frac{H+1}{2H^2}\right)^{H/(2H+1)}.$$

This factor is strictly larger than 1 for $H < 1$ and equal to 1 at $H = 1$.

A.2 Two-point lower bound used in proposition 2.11

The structural lower bound above is the exponent-level consequence of the exact Gaussian two-point calculation. For the ramp pair

$$\mu_{t-j}^{\pm,h} = \pm\zeta(h-j)_+^H,$$

the present-time separation is $2\zeta h^H$, so any estimator incurs risk at least of order $L_h^{\mathrm{ramp}}(\sigma,\zeta)$. Minimizing over h and using the asymptotic just proved gives the lower scale

$$\sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}$$

and therefore the horizon exponent $(2H+1)^{-1}$ for the root-n regime.

B Finite-Sample and Operational Diagnostics

This appendix gives finite-sample diagnostics for the theorem line and the remaining operational frontier. These diagnostics are not substitutes for theorem statements.

B.1 Finite-Sample Geometry Diagnostic

For the finite-sample term, we estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log–log slope in n. We also probe the fixed-span one-dimensional triangular window directly by comparing a triangular-array sample, with one draw from each drifted component, against an i.i.d. sample from the same window mixture. In that diagnostic the within-window span ζn^H is held fixed. Finally, we probe the Sinkhorn diagnostic at a few fixed values of ε.

Table 3: Empirical scaling diagnostic for the finite-sample term. The bounded-support i.i.d. W_2 experiment is close to the expected $n^{-1/2}$ rate in low dimension, the corresponding triangular-array diagnostic compares one sample from each drifted component against an i.i.d. sample from the window mixture, and the Sinkhorn diagnostic is reported at fixed ε. These rows are numerical support for the operational finite-sample discussion rather than theorem statements.

Setting	Estimated slope	Comment
W_2, $d = 1$	-0.48	close to $-1/2$
W_2, $d = 2$	-0.41	slower decay
W_2, $d = 4$	-0.25	slower decay
W_2, $d = 5$	-0.24	slower decay
W_2, i.i.d. mixture, $d = 1$, $H = 0.5$	-0.50	fixed span $\zeta n^H = 0.25$
W_2, triangular, $d = 1$, $H = 0.5$	-0.45	fixed span $\zeta n^H = 0.25$
W_2, i.i.d. mixture, $d = 1$, $H = 1.0$	-0.47	fixed span $\zeta n^H = 0.25$
W_2, triangular, $d = 1$, $H = 1.0$	-0.41	fixed span $\zeta n^H = 0.25$
Sinkhorn $\varepsilon = 0.50$	-0.21	stability constant changes
Sinkhorn $\varepsilon = 0.10$	-0.53	stability constant changes
Sinkhorn $\varepsilon = 0.05$	-0.67	stability constant changes

In low dimension, the raw i.i.d. Wasserstein term is compatible with a root-n rate, whereas higher-dimensional raw Wasserstein slows down. In the one-dimensional fixed-span triangular-array diagnostic, the recovered slopes stay near the corresponding i.i.d. mixture baseline, which is consistent with the proof-model and design-effect reading used in the 1D proof-model proposition.

B.2 Bahadur Remainder Diagnostics

Two deformation families make the proof-model remainder assumption concrete. The first is a smooth fixed-support monotone warp class $g_\delta(x) = x + \delta x(1-x)$ with $|\delta| \le \Delta < 1$. On the tested spans 0.2 and 0.4, the integrated remainder decays at rates between 1.40 and 1.54, and the scaled quantity $n \int_0^1 R_n(u)^2 du$ still decays at rates between 0.40 and 0.54. The second is the fixed-span triangular-array uniform-kernel model used in the proof line. There the integrated residual rate is about 1.43 for the triangular sample and 1.49 for the i.i.d. mixture benchmark, while the residual-to-MSE ratio drops from about 0.38 at $n = 25$ to about 0.11 at $n = 400$.

These diagnostics locate the role of drift span explicitly. Within the tested fixed-span range, the remainder stays lower-order and the empirical rate remains well above the n^{-1} threshold needed for item (iv). Rougher kernels and growing-span experiments remain less regular, which is why the analysis keeps the general triangular-array verification problem open.

B.3 Extended and Operational Regimes

The extended rows compare triangular and i.i.d. mixture rates on embedded low-intrinsic-dimensional support across span. The operational rows compare fixed-ε Sinkhorn rates across regularization values on embedded low-intrinsic-dimensional support in ambient dimension eight.

Table 4: Representative rows for the extended and operational finite-sample regimes. The extended rows compare triangular and i.i.d. mixture rates across span, while the operational rows compare fixed-ϵ Sinkhorn across ϵ on embedded low-intrinsic-dimensional support.

Regime	Setting	Parameter	Rate exponent a
extended	triangular ambient d=8, intrinsic k=1, span=0.10	0.10	0.4652
extended	iid mixture ambient d=8, intrinsic k=1, span=0.10	0.10	0.5722
extended	triangular ambient d=8, intrinsic k=1, span=0.25	0.25	0.4737
extended	iid mixture ambient d=8, intrinsic k=1, span=0.25	0.25	0.5588
extended	triangular ambient d=8, intrinsic k=1, span=0.50	0.50	0.4792
extended	iid mixture ambient d=8, intrinsic k=1, span=0.50	0.50	0.5425
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.50	0.50	0.5599
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.50	0.50	0.5706
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.20	0.20	0.5466
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.20	0.20	0.6150
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.10	0.10	0.4646
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.10	0.10	0.5060
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.05	0.05	0.4339
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.05	0.05	0.5124

The extended-regime rows indicate that low intrinsic dimension preserves a rate near the root-n regime without a dramatic triangular-versus-benchmark gap. The operational-regime rows record empirical evidence relevant to the fixed- ε Sinkhorn setting in definition 2.16: together with exact support-complexity inheritance and the dual threshold $\alpha > k/2$, they align with the embedded bandwise statement. A full proof of triangular-array inheritance would close this loop.

Two complementary diagnostics sharpen the same boundary. First, direct probes of the centered Sinkhorn scaling Jacobian on the embedded fixed-span model show that bands approaching zero are the numerically unstable region, whereas on a moderate band such as $[0.2, 0.8]$ the cross-coupling block remains contractive and the self-coupling inverse norms appear to plateau with sample size. Second, on the supported synthetic frontier grid, intrinsic-dimension estimates predict the observed stable-band cutoff better than ambient dimension. These diagnostics identify a useful regime statistic and isolate the band where the remaining instability is concentrated. They do not upgrade the embedded fixed- ε statement from conjecture to theorem.

The calibrated summary now separates three ingredients on the same embedded grid: the empirically stable band, the theorem-ready band obtained from the structural certificate, and the self-coupling proxy on the calibrated pairs where that proxy is currently informative. On $(8, 2)$ and $(12, 2)$, the empirical and theorem-ready bands both reach 0.5, and the two self-coupling blocks keep positive spectral gap, worst spectral radius below 0.97, and largest-sample mean inverse norm below 3.0.

Pair	Emp. $\varepsilon_{\max}$	Thm.-ready $\varepsilon_{\max}$	Min iid a	Min tri. a	Self
$(8, 1)$	0.5	0.5	0.488	0.458	–
$(8, 2)$	0.5	0.5	0.455	0.419	yes
$(12, 1)$	0.2	0.2	0.501	0.474	–
$(12, 2)$	0.5	0.5	0.428	0.493	yes

Table 5: Combined moderate-band Sinkhorn summary on the calibrated embedded grid. The table reports the empirical stable band, the theorem-ready band, the minimum recovered carrier exponents inside the theorem-ready band, and whether both self-coupling blocks satisfy the current spectral-radius and inverse-norm proxy thresholds.

B.4 Smooth Multivariate Transfer Diagnostics

The multivariate transfer diagnostic uses a drifting three-dimensional mean vector and a smooth log-sum-exp functional. The finite-sample error of the window estimator, the drift-induced bias of the window target, and the linearization residual are tracked separately across window lengths. The target behavior is the same as in the abstract multivariate transfer lemma: the finite-sample piece decreases with window length, the drift piece increases, and the linearization residual remains lower order than the finite-sample term on the tested windows.

Window	Tot.	Fin.	Drift	Resid.
8	0.2009	0.1988	0.0165	0.0239
16	0.1513	0.1542	0.0355	0.0162
32	0.1373	0.1363	0.0735	0.0066
64	0.1492	0.0807	0.1503	0.0051

Table 6: Smooth multivariate transfer diagnostic under triangular-array drift. The finite-sample term decreases with window length, the drift term increases, the total error has an interior minimum, and the linearization residual stays lower order across the tested windows.

B.5 Robustness Summaries

The real-stream, validity-gap, and online-controller sweeps are summarized here in compact form. Their role is to delimit where the qualitative signatures remain stable on the tested grids.

Variable	Runs	Interior	Best-window range	Edge/best range
`nswprice`	9	9/9	14–435	1.31–1.68
`nswdemand`	9	9/9	25–327	1.15–3.24

Table 7: ELEC2 robustness summary across anchor sizes 24, 48, and 96 and segment starts 0, 8000, and 16000, with segment length 4000. The interior-optimum count records how often the best window stays away from the search boundary; the edge/best range reports the ratio between the worse boundary error and the optimum.

Detector input	Detector	Detection rate	Mean-gap range	Excess-area range
Observation	ADWIN	1.00	429–620	7.41–20.41
Observation	Page–Hinkley	1.00	205–345	3.55–8.20
Abs. residual	ADWIN	0.00	—	—
Abs. residual	Page–Hinkley	0.33–0.92	994–1665	16.94–151.24

Table 8: Validity-detection robustness summary across operating windows 160, 220, and 300 and the reported threshold grid. Mean-gap and excess-area ranges are computed over the successful runs in each detector/input family.

Profile	Seed	Plug-in/oracle	Activity/oracle	Structural/oracle	Adaptive/oracle
Default	0	4.14	1.41	1.33	1.30
Default	1	4.17	1.33	1.38	1.31
Rough	0	2.50	1.65	1.34	1.49
Rough	1	2.18	1.40	1.26	1.30

Table 9: Reduced online-controller sweep with profiles `default` and `rough`, seeds 0 and 1, and maximum one-step window-index jumps. Entries report mean absolute error relative to the oracle controller.

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